

RESEARCH ARTICLE

EFFResNet-ViT: A Fusion-Based Convolutional and Vision Transformer Model for Explainable Medical Image Classification

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ABSTRACT The rapid advancement of medical imaging technologies requires the development of advanced, automated, and interpretable diagnostic tools for clinical decision-making. Although convolutional neural networks (CNNs) have shown significant promise in medical image analysis, they have limitations in capturing the global context and lack interpretability, thereby hindering their clinical adoption. This study presents EFFResNet-ViT, a novel hybrid deep learning (DL) model designed to address these challenges by combining EfficientNet-B0 and ResNet-50 CNN backbones with a vision transformer (ViT) module. The proposed architecture employs a feature fusion strategy to integrate the local feature extraction strengths of CNNs with the global dependency modeling capabilities of transformers. The extracted features are further refined through a post-transformer CNN and a global average pooling layer to enhance the classification performance. To improve interpretability, EFFResNet-ViT incorporates Grad-CAM visualization techniques to highlight regions contributing to classification decisions and employs t-distributed stochastic neighbor embedding for feature space analysis, providing insights into class separability. The proposed model was evaluated on two benchmark datasets: brain tumor (BT) CE-MRI for BT classification and a retinal image dataset for ophthalmological diagnosis. EFFResNet-ViT achieved state-of-the-art performance, with accuracies of 99.31% and 92.54% on the BT CE-MRI and retinal datasets, respectively. Comparative analyses demonstrate the superior classification performance and interpretability of EFFResNet-ViT over existing ViT and CNN-based hybrid models. The explainable design of EFFResNet-ViT addresses the critical need for transparency in artificial intelligence-driven medical diagnostics, facilitating its potential integration into clinical workflows to improve decision-making and patient outcomes.

INDEX TERMS EfficientNet-B0, ResNet-50, vision transformer, model explainability, t-distributed stochastic neighbor embedding.

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I. INTRODUCTION

Medical image classification is critical to modern healthcare, enabling accurate and efficient diagnosis in various conditions. This process involves the analysis and categorization of

medical images based on various factors, including clinical information and imaging modalities. The ability to classify medical images reliably has profound implications for patient care because it facilitates faster and more precise clinical evaluations [1].

Recently, convolutional neural networks (CNNs) have revolutionized the field of medical image analysis. These sophisticated deep learning (DL) models have become the cornerstone of numerous computer-aided diagnostic (CAD) systems, marking the beginning of a new era of artificial intelligence (AI) in clinical settings [2]. CNNs have demonstrated an impressive ability to learn complex patterns from large medical image datasets. These advanced systems can often diagnose conditions with a precision that matches or exceeds that of experienced doctors [3]. CNN-based models have achieved performances comparable to those of human experts in tasks such as detecting pneumonia from chest X-rays and classifying optical coherence tomography (OCT) images [4].

CNNs have revolutionized medical image analysis by automatically extracting hierarchical features from images, thereby eliminating the need for manual feature engineering. This capability makes them particularly effective for tasks such as tumor detection in MRI scans [5], [6], X-ray image analysis [7], [8], identification of eye diseases [9], [10], polyp analysis [11], and classification of skin conditions [12], [13]. The success of CNNs in these diverse applications underscores their potential to transform medical image analysis into various specialties.

Despite their impressive achievements, CNNs exhibit certain limitations. A major drawback is their inherent locality bias, which makes it challenging for them to capture long-range dependencies in visual data. Furthermore, the texture, shape, and size of many organs vary significantly among individuals, complicating the accurate analysis of medical data [14], [15]. Therefore, the extraction of robust feature representations that can model long-range dependencies across domains is crucial for practical medical image analysis.

Researchers have turned to transformer architectures to address these challenges, which were initially developed for language processing tasks [16]. With their self-attention mechanisms, transformers have shown a remarkable potential to model long-range relationships in visual data [17]. By allowing each element of an input to interact with every other element, transformers can capture the overall context more effectively than CNNs. Recent studies have demonstrated the superiority of transformer-based models in various medical image analysis tasks, outperforming traditional CNN architectures in terms of accuracy and generalization [18], [19], [20], [21]. A comprehensive review of 36 studies indicated that transformer-based models, particularly the vision transformer (ViT), exhibit significant potential in diverse medical imaging tasks, showing superior performance compared to conventional CNN models [22].

The self-attention mechanism in transformers enables these models to dynamically weigh the importance of different image regions, making them particularly good at understanding the complex spatial relationships often found in medical images. This ability is beneficial for organ segmentation tasks, where understanding the overall structure and local features is important.

However, the use of transformer models in medical image analysis is challenging. A major problem is that they require considerable amounts of training data to exploit their potential fully. Unlike CNNs, which perform well with smaller datasets because of their inherent biases, transformers typically require large-scale datasets for effective learning. This poses a challenge in the medical domain, where high-quality labeled datasets are often complex owing to privacy issues, high cost of expert labeling, and time-intensive data collection.

With the evolution of this field, researchers are exploring hybrid approaches that combine the strengths of CNN with those of transformers. In this study, we propose EFFResNet-ViT, a novel fusion-based model that integrates the local feature extraction capabilities of EfficientNet-B0 and ResNet-50 with the global context modeling of ViT for explainable medical image classification. The main contributions of this study are as follows.

- 1) We propose EFFResNet-ViT, a novel hybrid DL model that combines CNN backbones, EfficientNet-B0 [23] and ResNet-50 [24], with a ViT module, integrating local feature extraction strengths with global dependency modeling capabilities for improved medical image classification.
- 2) The extracted features are further refined through post-transformer CNN and global average pooling (GAP) layers, enhancing the ability of the model to capture fine-grained details and aggregate spatial information, improving classification performance and generalization capabilities.
- 3) The interpretability of the model is improved using grad-CAM [25] visualization techniques to highlight regions that contribute to classification decisions and employing t-distributed stochastic neighbor embedding (t-SNE) [26] for feature space analysis, providing insights into class separability and addressing the critical need for transparency in AI-driven medical diagnostics.

The remainder of this paper is organized as follows. Section II presents the related work. Section III examines the proposed methods. Section IV explains and analyzes the experiments, including the data preprocessing and metric evaluation. Section V presents a detailed explanation of the results and discussion for both datasets, including the model explainability. Section VI compares the performance of the proposed model with that of the state-of-the-art (SOTA) models. Finally, Section VII concludes the study.

II. RELATED WORK

A. CNN-BASED METHODS

CNNs have significantly affected medical imaging by utilizing their ability to learn elaborate and nuanced representations using data-driven methodologies. Jia et al. [27] developed a solution to address gradient vanishing difficulties by dynamically integrating information from different layers of MobileNet and ResNets. This technique successfully conserved critical information throughout the network design and improved performance. The study presented in [28] focused on improving input data by removing boundary regions and filling gaps around the lungs. Based on the ResNet-50 framework, they used a feature pyramid network module and a multilayer perceptron (MLP) to predict. This study [29] presented C-Net, an automated CNN architecture to analyze histopathology images. In the C-Net framework, several CNN were used to classify the images as malignant or benign. In [30], an adaptive momentum optimizer was employed to train CNNs for medical image classification. This dynamic momentum algorithm automatically adjusts the momentum rate based on errors, eliminating the need for manual hyperparameter tuning. To achieve accurate brain disorder diagnoses, [31] applied CAD to magnetic resonance imaging (MRI) scans. They introduced a multi-step preprocessing technique to improve the quality of MRI images and a novel CNN architecture for identifying malignant gliomas, meningiomas, pituitary tumors, and normal brain regions.

B. TRANSFORMER-BASED METHODS

Attention-based transformer models have revolutionized DL by capturing long-range relationships and learning sophisticated feature representations. ViT [20] designs have emerged as a significant development, replacing convolutions with image patch sequences and achieving SOTA performance across several computer vision (CV) tasks. The study presented in [32] introduced a unique multi-view network, entirely based on transformers, for the classification of mammographic images. This method employed attention mechanism that effectively fused data from various views to enhance the analysis. Furthermore, a token-based approach was introduced to capture essential information for accurate classification during the final prediction stage. In [33], a transformer-based multimodal DL framework was developed to combine multiple data types for analyzing lesion tumors, incorporating both clinical and dermoscopic images, as well as patient-specific information. This method facilitated more detailed and accurate skin tumor detection and classification. Meanwhile [34], a self-ensembling ViT (SEViT) was introduced to improve the ability of ViT models to handle adversarial attacks. SEViT strengthened the model by using the stability of early block features to resist adversarial changes and combining results from multiple classifiers with the final ViT output. This approach was tested on chest X-ray and funduscopy data, showing it can handle different types of adversarial attacks in a gray-box environment.

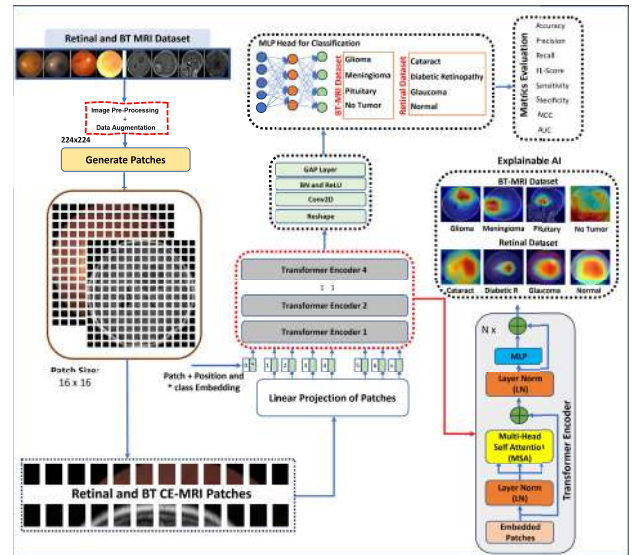


FIGURE 1. The proposed ViT model processes the input images as patches through transformer encoders with self-attention, followed by a post-transformer convolutional layer and a custom MLP head for classification. Grad-CAM is used to generate heatmap visualizations, highlighting key image regions to enhance model explainability.

C. HYBRID METHODS

Hybrid models for medical image categorization have emerged as viable techniques for enhancing the accuracy and resilience of diagnostic systems. Several studies have proposed various hybrid designs to exploit the strengths of CNNs and transformers. One such approach [35] used a hybrid CNN model incorporated a ResNet 152 layer to enhance feature extraction. Another study [36] introduced the PHTrans model, which combines parallel CNNs and transformers to effectively capture both global and local features, thereby enhancing segmentation performance. In [37], the authors combined DHRNet and a hybrid transformer to extract local and global features, aiding in exploring long-range dependencies. A different hybrid approach [38] used a combination of CNNs and local binary patterns (LBP) for image classification. In this approach, multiple classifiers collaborated by calculating the Euclidean distance between LBP feature vectors and the confidence levels of CNN features, which were classified by support vector machines (SVMs). These values served as inputs to a multi-layer perceptron (MLP) classifier. The results from these classifiers were then combined to form a final voting ensemble. Similarly, the study in [39] introduced CTCNet, which combined Swin transformers with residual CNNs using a cross-domain fusion block to effectively merge features. MedViT [40] presented a hybrid CNN-transformer architecture designed for medical image diagnosis. This approach addresses issues related to adversarial attacks and the reliability of deep medical systems by merging the local feature extraction power of CNNs with the global connectivity of transformers. Additionally, they introduced a more efficient convolution operation to reduce computational complexity and methods to improve model robustness and

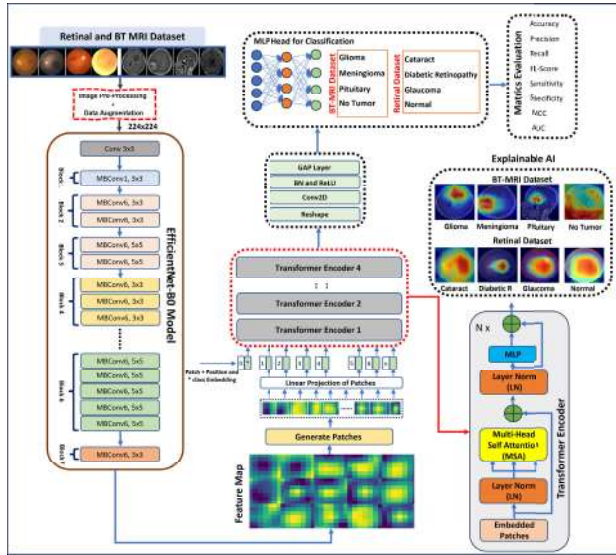


FIGURE 2. The proposed hybrid EfficientNetB0-ViT model for multiclass medical image classification uses EfficientNet-B0 for local feature extraction and ViT for global context modeling. A post-transformer convolutional layer is followed by a custom MLP classification head. Grad-CAM is employed to provide explainability through heatmap visualizations.

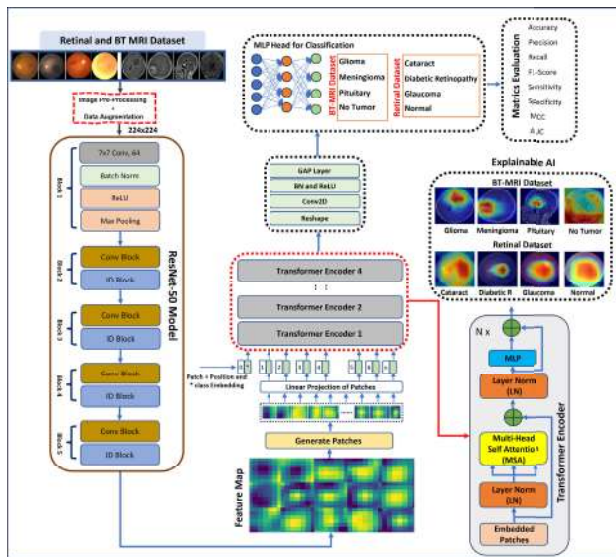


FIGURE 3. The proposed hybrid ResNet50-ViT model for multiclass medical image classification uses ResNet-50 for local feature extraction and ViT for capturing the global context, followed by a post-transformer convolutional layer and a custom MLP classification head. Grad-CAM is used to provide explainability through heatmap visualization.

decision boundary smoothness. In [41], Jang and Hwang proposed a 3D medical image classifier using the multiplane and multislice transformer (M3T) network to accurately detect Alzheimer's disease in 3D MRI scans. The M3T network combined 3D CNN, 2D CNN, and a transformer, leveraging their strengths in learning representations and attention mechanisms. The 3D CNN used inductive bias to detect local anomalies, whereas the transformer did not use inductive bias to detect abnormalities across a broader region.

TABLE 1. Details of the BT CE-MRI image dataset.

Tumor Class	Training Images	Validation Images	Testing Images
Glioma Tumor	1057	264	300
Meningioma Tumor	1072	267	306
Pituitary Tumor	1166	291	300
No Tumor	1276	319	405
Total Images	4571	1141	1311

TABLE 2. Details of the retinal image dataset.

Retinal Condition	Training Images	Validation Images	Testing Images
Cataract	726	156	156
Diabetic Retinopathy	768	165	165
Glaucoma	705	151	151
Normal	752	161	161
Total Images	2951	633	633

TABLE 3. Hyperparameter values for the proposed model.

Parameters	Values
Optimizer	Adam
Image size	$224 \times 224 \times 3$
Patch size	16
Number of Transformer blocks	4
Hidden dimension in Transformer	128
MLP dimension	64
Number of heads	4
Activation function	ReLU
Loss function	Categorical cross-entropy
Batch size	32
number of patches	196
Patience	5
Monitor	Validation loss
Weight decay	0.001
Optimizer	Adam
Learning rate	$1e-4$

III. METHODS

This study proposes the EFFResNet-ViT framework to classify and localize medical images precisely. Our architecture integrates the strengths of EfficientNet-B0 [23] and ResNet-50 [24] for local feature extraction combined with the global context modeling capabilities of ViT. This fusion approach enables EFFResNet-ViT to effectively capture fine-grained details and long-range dependencies in medical images.

A. ViT MODEL

The ViT [20] model is based on the transformer architecture, which was initially designed for natural language processing (NLP) and adopted for CV tasks [42]. ViT recognizes spatial hierarchy as a sequence of patches, unlike CNNs, which use convolutional layers. Consequently, it can capture

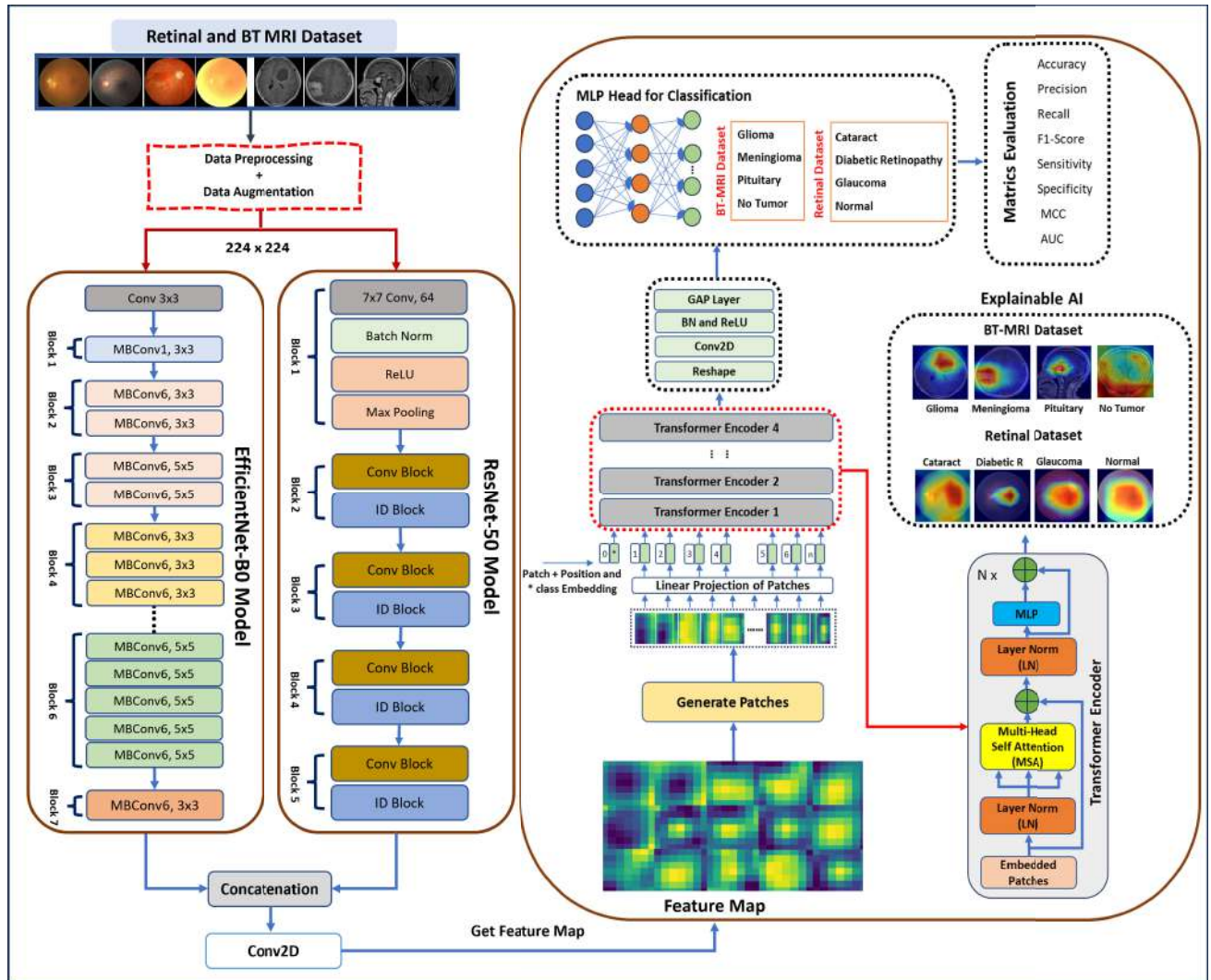


FIGURE 4. The proposed EFFResNet-ViT model framework combines CNNs (EfficientNet-B0 and ResNet-50) for local feature extraction with ViT for global feature modeling. The key components include patch embeddings, a transformer encoder, post-transformer convolutional layers, and a custom MLP head for classification. Grad-CAM visualizations enhance interpretability by highlighting critical regions relevant to tumor classification.

TABLE 4. Training and validation performance of the proposed models on the BT CE-MRI and retinal datasets.

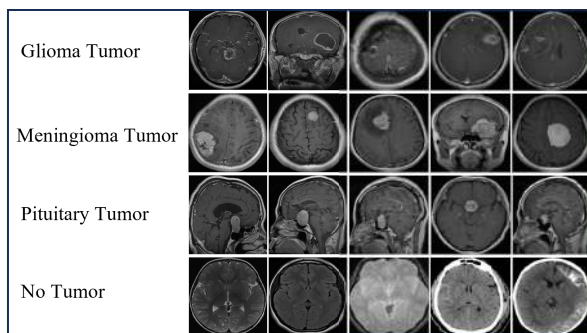
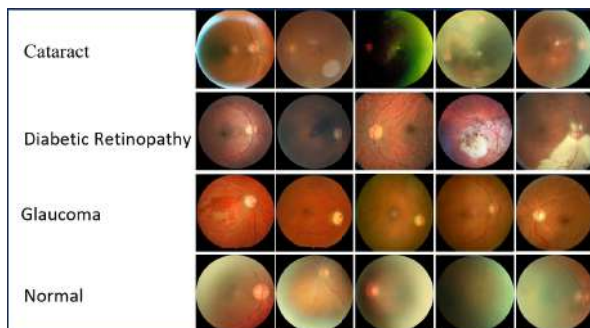
Model	Datasets	Train Accuracy	Validation Accuracy	Train Loss	Validation Loss
ViT	BT CE-MRI	0.8768	0.8071	0.3279	0.5295
	Retinal	0.9054	0.7392	0.2590	0.8262
EfficientNetB0-ViT	BT CE-MRI	0.9982	0.9614	0.0301	0.1961
	Retinal	0.9996	0.9274	0.0720	0.3402
ResNet50-ViT	BT CE-MRI	0.9831	0.9211	0.1419	0.3359
	Retinal	0.9988	0.9125	0.066	0.4502
EffResNet-ViT	BT CE-MRI	1.00	0.9921	0.0097	0.0467
	Retinal	0.9996	0.9229	0.0363	0.3284

global dependencies and contextual information across images. The model begins by dividing the input image into fixed-size, non-overlapping patches, utilizing a patch size

of 16×16 to balance detailed feature capture with computational efficiency. These patches are then flattened and linearly embedded in lower-dimensional vectors with an embedding

TABLE 5. Test performance of the proposed models on the BT CE-MRI and retinal datasets.

Model	Datasets	Accuracy	Precision	Recall	F1-Score	Sensitivity	Specificity	MCC	AUC
ViT	BT CE-MRI	0.8009	0.8031	0.8009	0.7949	0.8009	0.9326	0.7353	0.9549
	Retinal	0.7310	0.7313	0.7310	0.7308	0.7310	0.9108	0.6414	0.9130
EfficientNetB0-ViT	BT CE-MRI	0.9832	0.9832	0.9832	0.9830	0.9832	0.9945	0.9775	0.9995
	Retinal	0.9218	0.9216	0.9218	0.9214	0.9218	0.9741	0.8959	0.9862
ResNet50-ViT	BT CE-MRI	0.9558	0.9565	0.9558	0.9555	0.9558	0.9853	0.9410	0.9973
	Retinal	0.9005	0.8999	0.9005	0.9000	0.9005	0.9670	0.8673	0.9792
EffResNet-ViT	BT CE-MRI	0.9931	0.9926	0.9925	0.9926	0.9925	0.9978	0.9908	0.9991
	Retinal	0.9254	0.9260	0.9254	0.9255	0.9254	0.9752	0.9005	0.9882

**FIGURE 5.** Multiclass BT CE-MRI sample dataset.**FIGURE 6.** Sample Retinal image dataset.

dimension of 128, ensuring sufficient representation capacity for each patch. A learnable classification token is added to the sequence of embedded patches and positional embeddings are added to provide spatial information.

The core of ViT consists of several layers of transformer encoders, each containing multi-head self-attention (MHSA). Specifically, we utilize MHSA with 4 heads, which enables the model to attend to different aspects of the image simultaneously. Each head learns distinct patterns, enhancing the model's ability to capture richer relationships and discern complex features compared to a single-head approach. The choice of 4 attention heads provides a computationally efficient design while maintaining robust performance, consistent with empirical findings for baseline ViT models. The transformer layers also include feed-forward neural

networks (FFNN), layer normalization (LN), and residual connections [43]. This structure allows the model to capture complex relationships between different parts of an image in parallel. Using the self-attention mechanism, the model determines the relevant parts of the image for classification by modeling the input sequence into query, key, and value matrices. The output corresponding to the classification token, after transformer layers, serves as the global image representation. These transformer layers operate with a hidden dimension of 128, providing sufficient patch representation for effective processing.

To further refine feature extraction and integrate spatial relationships, the output sequence is reshaped into a 2D spatial structure and processed with a convolutional layer (Conv2D), followed by batch normalization (BN) and ReLU activation. This step addresses local spatial relationships potentially overlooked by global attention. The resulting features are then aggregated using GAP into a single vector. Finally, a classification head, typically an MLP with softmax activation, maps these pooled features to class probabilities. The model is trained to minimize categorical cross-entropy loss between predicted and ground-truth labels.

Through its patch-based representation and global self-attention, ViT effectively captures both local and global image patterns, demonstrating robust performance in image classification tasks, including medical image analysis. The overall architecture of the ViT model, including the hyperparameter choices mentioned in Table 3, is illustrated in Fig. 1.

B. EfficientNetB0-ViT

The EfficientNetB0-ViT model benefits from the complementary strengths of EfficientNet-B0 [23] and ViT [20], making it a robust choice for medical image classification tasks in which both local feature extraction and global context modeling are essential. The proposed EfficientNetB0-ViT architecture is shown in Fig. 2.

The model effectively handles complex medical image data by combining the EfficientNet-B0 [23] architecture, which is known for its efficient local feature extraction, with the capacity of ViT to capture long-range dependencies. The hybrid design allows the extraction of fine-grained

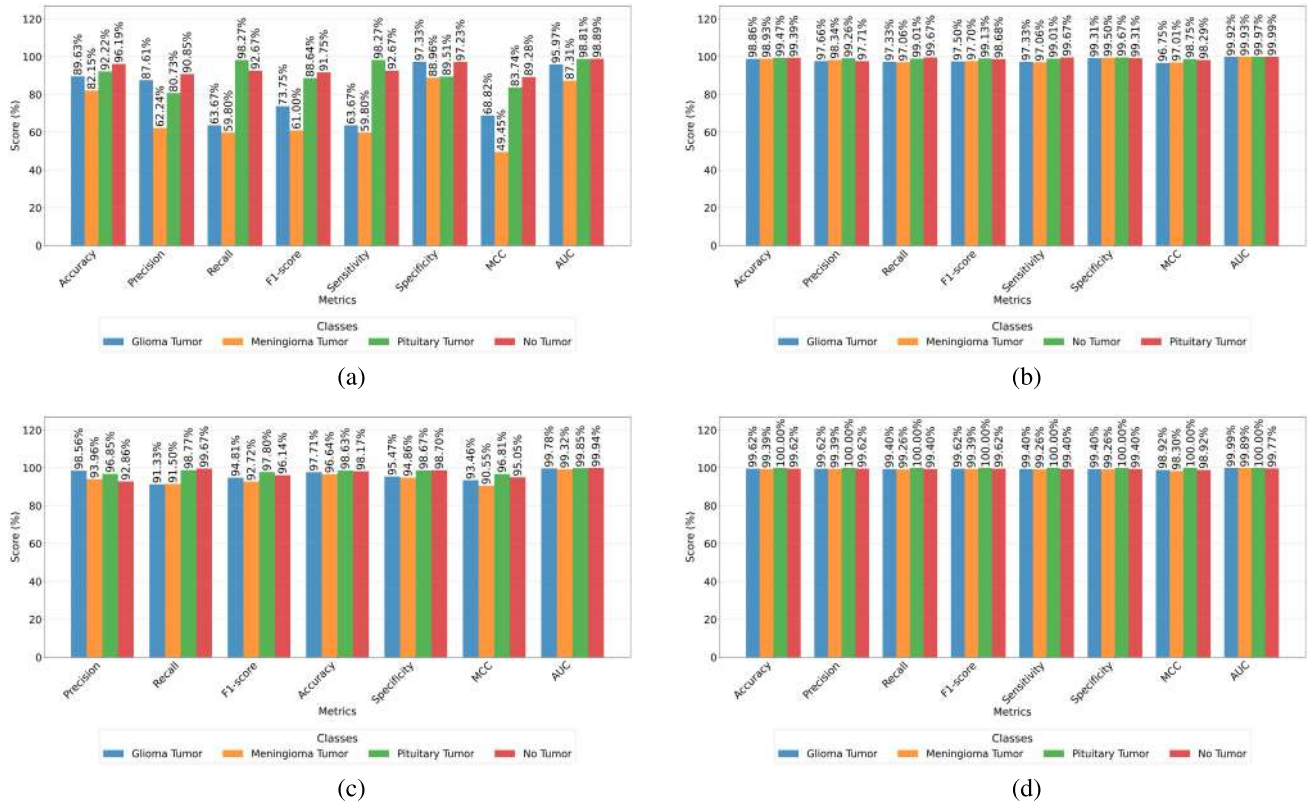


FIGURE 7. Class-wise performance of the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT models.

features through EfficientNet-B0 [23] while leveraging ViT's attention mechanisms to understand the global context. This is ideal for tasks such as medical image classification, where both local and global information contribute to an accurate diagnosis.

EfficientNet, proposed by Tan and Le [23], is a CNN that uniformly scales the depth, width, and resolution dimensions using a compound coefficient. This method enables EfficientNet to achieve SOTA accuracy while maintaining computational efficiency [44]. EfficientNet-B0 [23], which is the baseline model of the EfficientNet family, utilizes mobile inverted bottleneck convolution blocks that incorporating depthwise separable convolutions, Swish activation functions, and squeeze-and-excitation layers. These design choices allow EfficientNet-B0 [23] to outperform traditional models using fewer parameters, thereby offering an efficient way to extract local features from medical images.

The EfficientNetB0-ViT model begins with an input layer that accepts images of size $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels, respectively. EfficientNet-B0 [23] processes the input image and generates compact feature map $\mathbf{F} \in \mathbb{R}^{h \times w \times d}$, where h and w are the spatial dimensions of the feature map, and d is its depth. This feature extraction is performed using the EfficientNet-B0 [23] network, excluding its classification head, and fine-tuning the final block. The output feature map from Block_7 of EfficientNet-B0 [23] is mathematically

expressed as follows:

$$\mathbf{F} = \text{EffB0}_{\text{block}_7}(\mathbf{I}), \quad (1)$$

$\mathbf{F}$ denotes the feature map generated by Block_7 of EfficientNet-B0 [23] after processing input image $\mathbf{I}$. Feature map $\mathbf{F}$ is then passed through a patch embedding layer, implemented as a convolutional layer. The kernel size and stride of this convolutional operation are set equal to patch size P , which splits feature map $\mathbf{F}$ into non-overlapping patches and projects them into a latent space of dimension D . This results in a sequence of patch embeddings, $\mathbf{P}_{\text{emb}} \in \mathbb{R}^{N \times D}$, where $N = \frac{h \cdot w}{P^2}$ denotes the number of patches. The patch embedding process is defined as

$$\mathbf{P}_{\text{emb}} = \text{Conv2D}(D, \text{kernel size} = P, \text{stride} = P)(\mathbf{F}), \quad (2)$$

where $\mathbf{P}_{\text{emb}}$ represents the sequence of patch embeddings generated from feature map $\mathbf{F}$.

These embeddings are flattened into a 2D sequence where each patch is represented as a vector. Learnable positional embeddings are added to encode the spatial relationships within the original image. This combined sequence forms the input for the ViT encoder, as discussed in III-A.

By integrating EfficientNet-B0 [23] local feature extraction capabilities with ViT's global context modeling, the EfficientNetB0-ViT model captures fine-grained details and long-range dependencies in medical images. This renders it highly effective for tasks such as medical image classification.

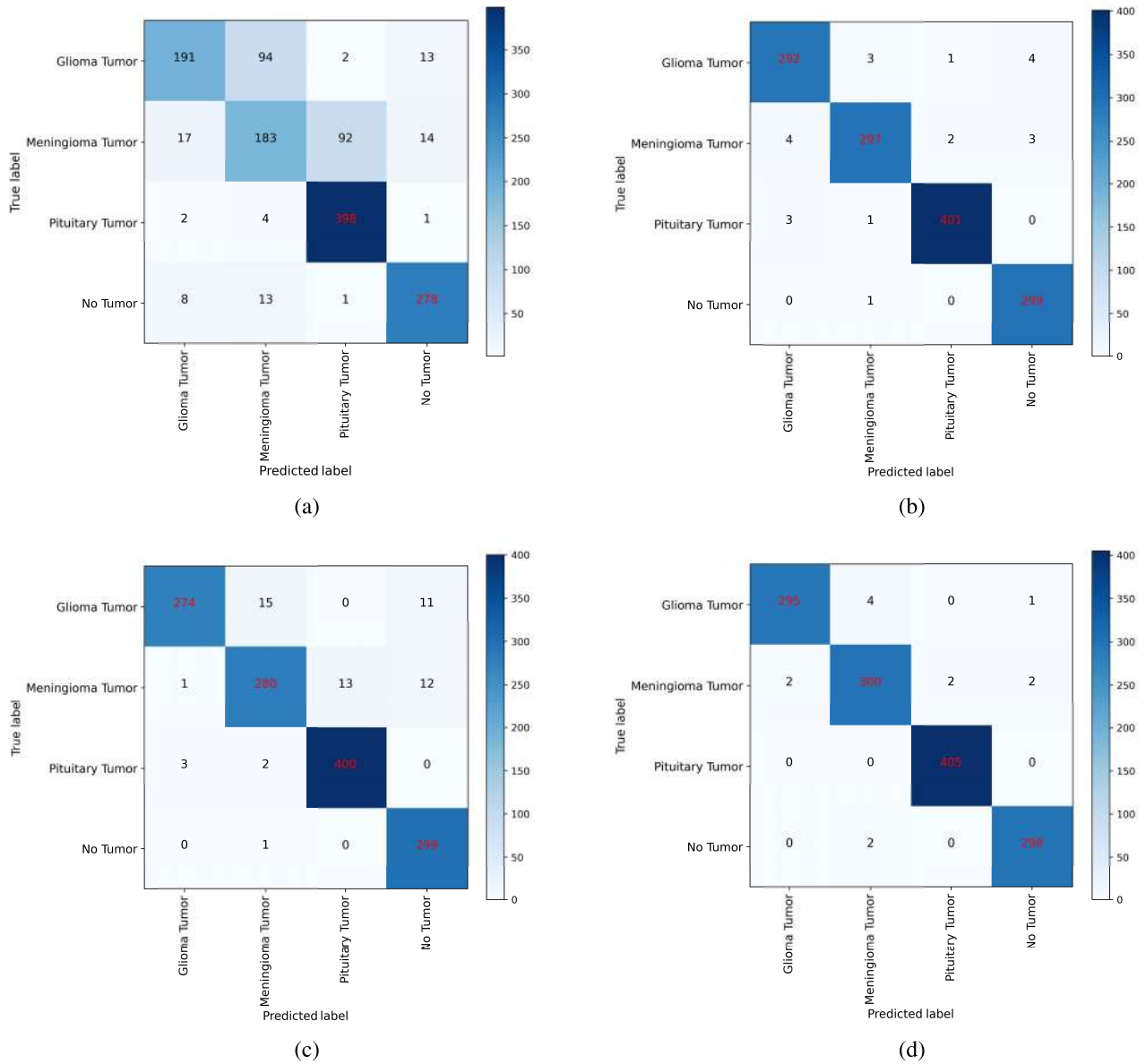


FIGURE 8. Confusion matrix of the proposed model on the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

C. ResNet50-ViT

The ResNet50-ViT model combines the advantages of ResNet-50 [24] and ViT [20] for medical image classification tasks. This hybrid architecture leverages ResNet-50 [24] as a feature extractor and integrates it with a ViT [20] model to enhance the global context understanding. The proposed ResNet50-ViT architecture is illustrated in Fig. 3.

ResNet50 [24], proposed by He et al. [24], is a deep residual network designed to facilitate the training of intense networks using residual connections, allowing gradients to flow more easily during backpropagation. The core idea of ResNet is to introduce identity shortcut connections that bypass one or more layers, making the network easier to optimize. These residual connections enable ResNet-50 [24]

to train networks with a depth of 50 layers without the vanishing gradient problem.

The ResNet-50 [24] architecture consists of convolutional layers followed by residual blocks. Each residual block involves convolutions with skip connections, which help the model learn low- and high-level features. ResNet-50 [24] has demonstrated superior performance in various image classification tasks, providing high accuracy and efficiency [45].

The ResNet50-ViT model begins with an input layer that accepts images of size $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote the height, width, and number of channels, respectively. In the ResNet50-ViT model, we modify the ResNet-50 architecture by removing its classification head

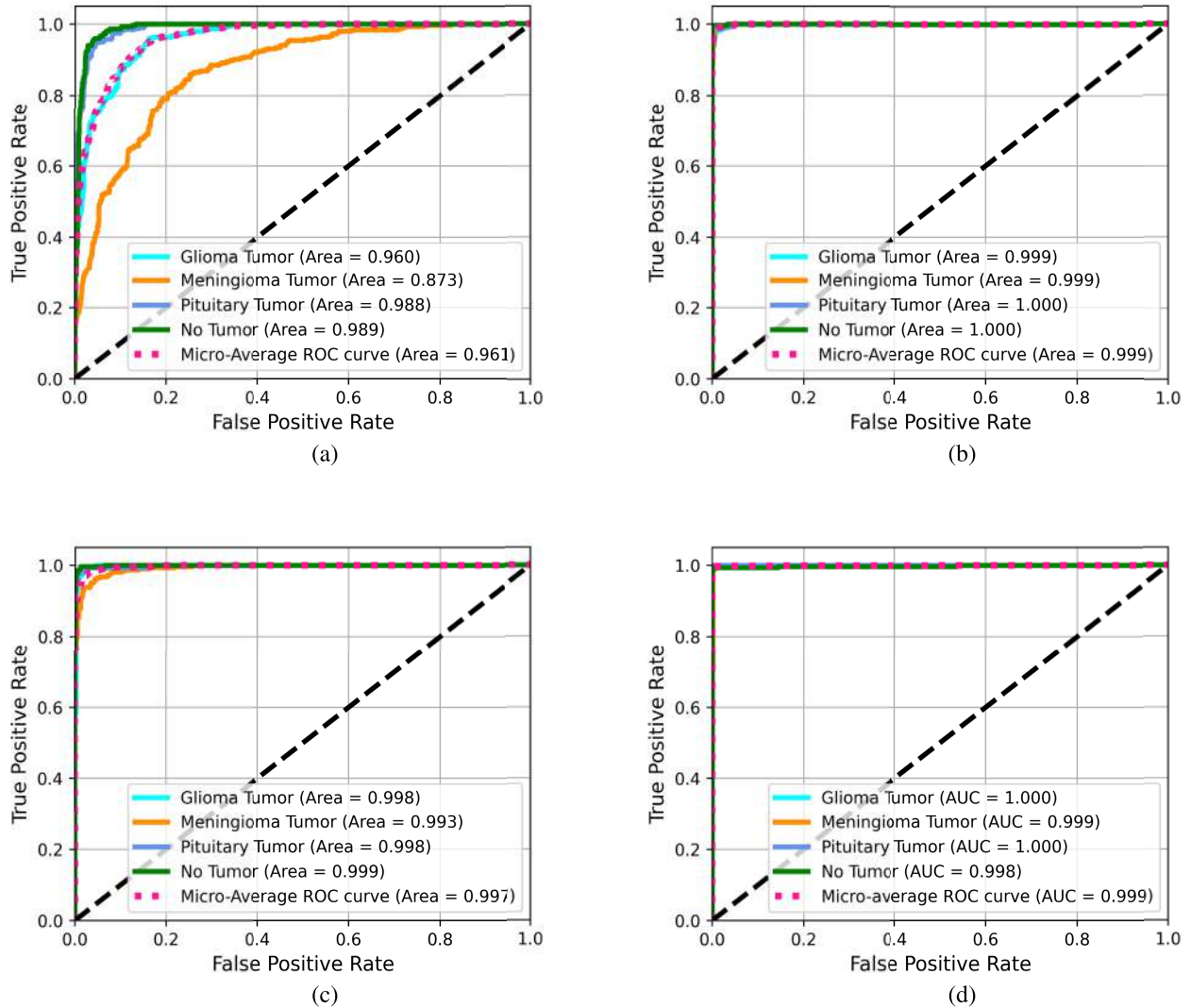


FIGURE 9. Receiver operating characteristic (ROC) curve of the proposed model on the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

and fine-tuning the final block. The output feature map from `Block_5` of ResNet-50 [24] is mathematically expressed as follows:

$$\mathbf{F} = \text{ResNet50}_{\text{block}_5}(\mathbf{I}), \quad (3)$$

$\mathbf{F}$ represents the feature map generated by `Block_5` of EfficientNet-B0 after processing input image $\mathbf{I}$. Feature map $\mathbf{F}$ is then passed through a patch embedding layer as discussed in III-B, and the resulting patch embeddings are processed by the ViT encoder as discussed in III-A.

This fusion approach enhances the ability of the model to understand local structures and global patterns, which is crucial for accurate medical image analysis.

D. PROPOSED EFFResNet-ViT MODEL

Recently, ViT [20] has emerged as a leading DL framework for image processing, surpassing traditional architectures

such as CNNs and recurrent neural networks in terms of both performance and computational efficiency, particularly in the field of medical imaging [40], [46]. ViT [20] utilizes a self-attention mechanism to effectively extract and model global features, making it particularly suitable for complex tasks in which understanding the spatial dependencies is critical. Given the complementary strengths of CNNs (for fine-grained local feature extraction) and ViT [20] (for capturing long-range dependencies), combining these architectures is a logical progression toward robust medical image analysis.

In this study, we propose EFFResNet-ViT, a hybrid model that integrates EfficientNet-B0 [23] and ResNet-50 [24] with a ViT module for multiclass medical image classification and localization. EfficientNet-B0 [23] and ResNet-50 [24] are fine-tuned by unfreezing their last blocks and removing their classification heads to enhance their adaptability to the biomedical domain.

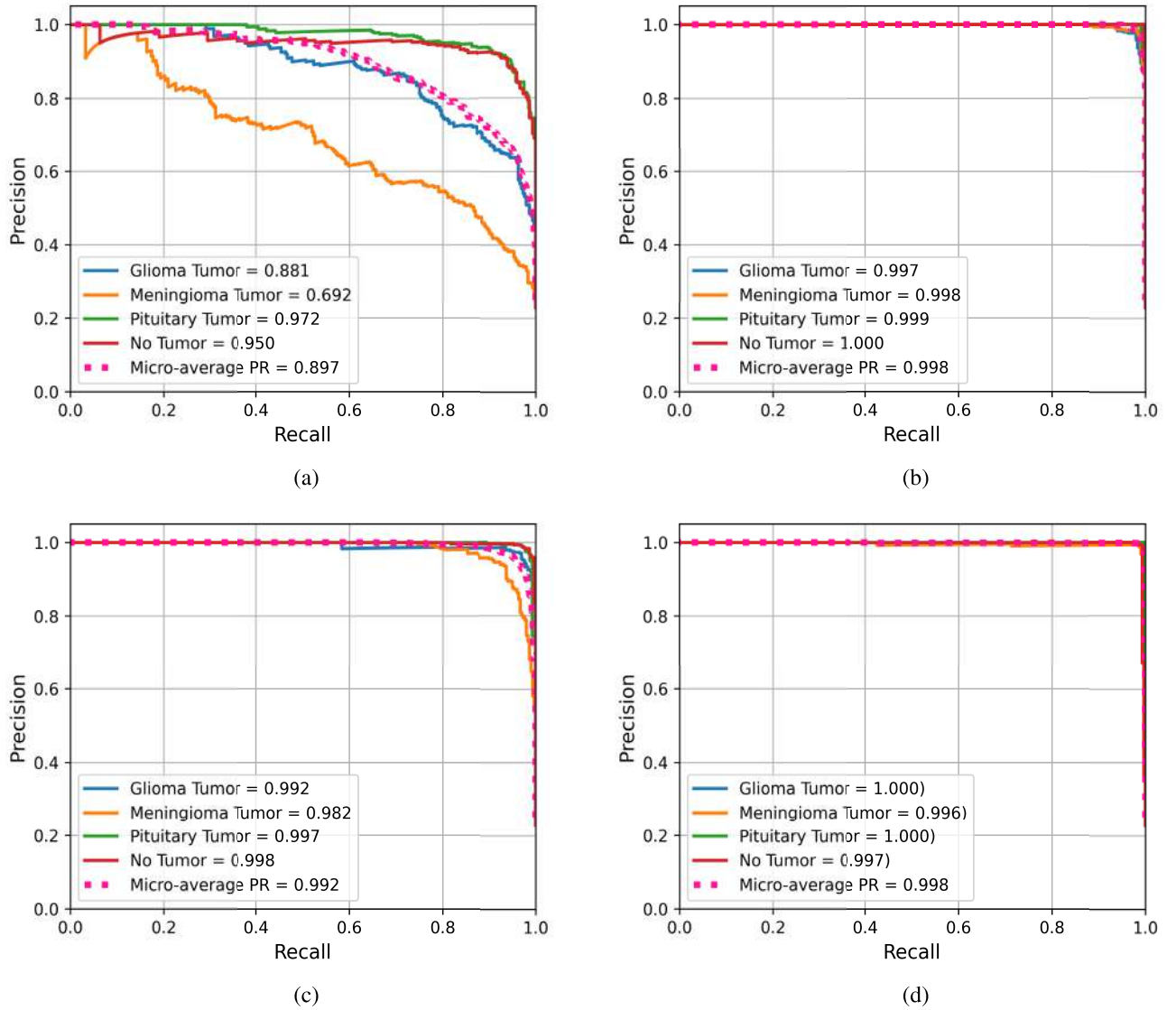


FIGURE 10. Precision-recall curve of the proposed models on the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

Input image $X \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and channels, respectively, is passed through two pretrained CNN models: EfficientNet-B0 [23] and ResNet-50 [24]. The local features are extracted as follows:

$$F_{\text{Eff}} = f_{\text{EffNet}}(X), \quad F_{\text{Res}} = f_{\text{ResNet}}(X) \quad (4)$$

where $f_{\text{EffNet}}(\cdot)$ and $f_{\text{ResNet}}(\cdot)$ represent the feature extraction functions of EfficientNet-B0 and ResNet-50, respectively.

The extracted features are concatenated to form a comprehensive feature representation.

$$F_C = \text{Concat}(F_{\text{Eff}}, F_{\text{Res}}) \quad (5)$$

The combined features, F_C , are then passed through a convolutional layer to generate patch embeddings (P_{emb}). Specifically, F_C is divided into non-overlapping patches of

size $P \times P$, and a convolution operation with a kernel size equal to patch size p transforms the combined features into hidden vectors of dimension D .

The combined features, F_C , are subsequently passed through a patch embedding layer, which is implemented as a convolutional layer. The kernel size and stride of this convolutional operation are set equal to patch size P , which splits feature map $\mathbf{F}$ into nonoverlapping patches and projects them into a latent space of dimension D . This results in a sequence of patch embeddings $\mathbf{P}_{\text{emb}} \in \mathbb{R}^{N \times D}$, where $N = \frac{h \cdot w}{P^2}$ denotes the number of patches. The patch embedding process is defined as:

$$\mathbf{P}_{\text{emb}} = \text{Conv2D}(D, \text{kernel size} = P, \text{stride} = P)(\mathbf{F}_C), \quad (6)$$

where $\mathbf{P}_{\text{emb}}$ represents the sequence of patch embeddings generated from feature map $\mathbf{F}_C$.

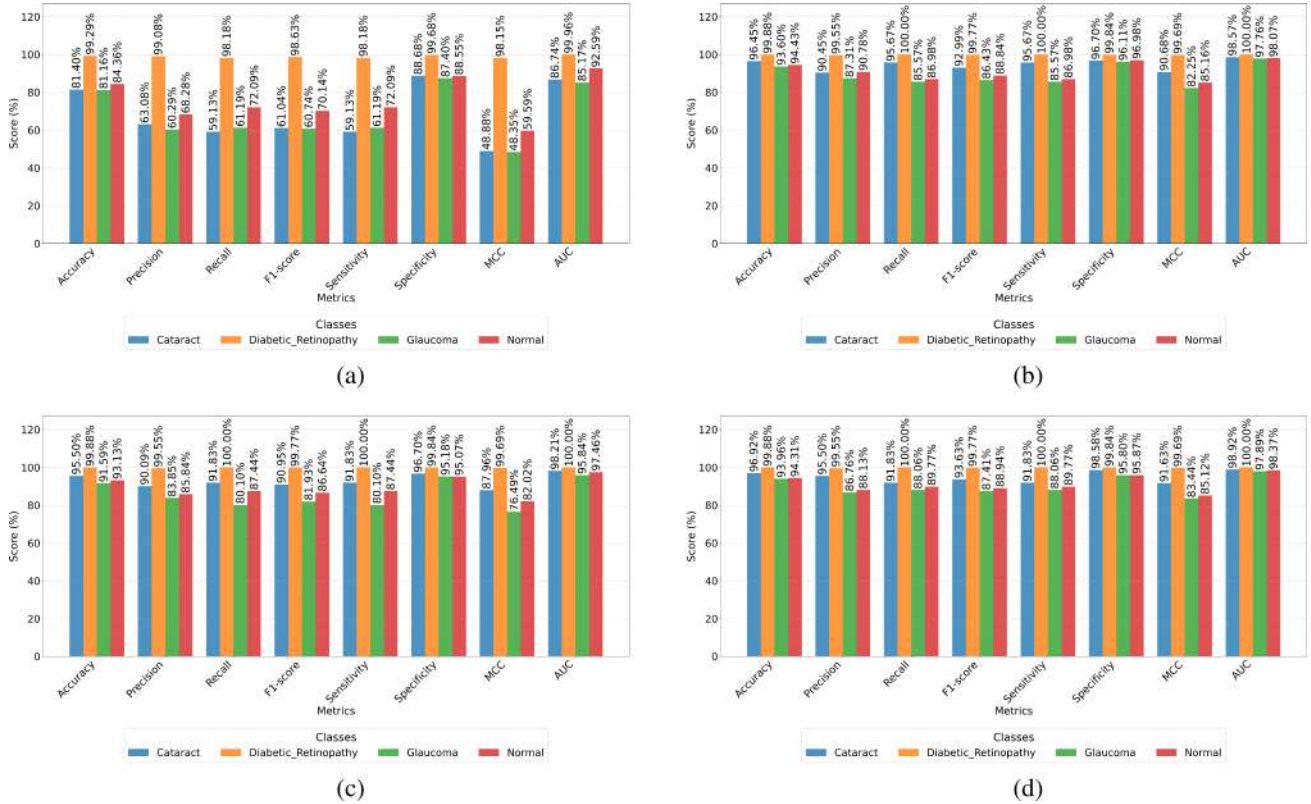


FIGURE 11. Class-wise performance of the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT models.

Each patch is flattened and reshaped into a sequence of shapes (N, D) , where N denotes the total number of patches.

$$N = \frac{H' \times W'}{p^2} \quad (7)$$

where H' and W' are the spatial dimensions of F_C after feature extraction.

Learnable positional embedding E_{pos} is added to P_{emb} to preserve spatial information.

$$Z_0 = P_{emb} + E_{pos} \quad (8)$$

A class token (x_{class}) is introduced to aggregate global information for classification. This token is concatenated with the patch embeddings to form the following input sequence for the transformer encoder:

$$Z'_0 = [x_{class}, Z_0] \quad (9)$$

The transformer encoder comprises multiple layers with an MHSA mechanism and an MLP block. Both components employ residual connections and LN for training stability. The MHSA block computes attention scores using query Q , key K , and value V matrices

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (10)$$

The transformer layer processes the input as follows:

$$z'_l = \text{LN}(z_{l-1} + \text{MHSA}(\text{LN}(z_{l-1}))) \quad (11)$$

$$z_l = \text{LN}(z'_l + \text{MLP}(\text{LN}(z'_l))) \quad (12)$$

After passing through L transformer layers, the output corresponding to the class token is used as the global representation of the image. To integrate the spatial relationships and enhance feature extraction, the output sequence is reshaped back into a 2D spatial structure as follows:

$$\mathbf{X}_{\text{reshaped}} \in \mathbb{R}^{N \times N \times D}. \quad (13)$$

A convolutional layer (Conv2D) is then applied, followed by BN and ReLU activation. This layer helps capture local spatial relationships that the global attention mechanism may have missed. The BN and ReLU layers further stabilize the training process and introduce nonlinearity. The output is then passed through GAP, which reduces the spatial dimensions to a single vector. This operation can be expressed as

$$\mathbf{X}_{\text{conv}} = \text{ReLU}(\text{BN}(\text{Conv2D}(\mathbf{X}_{\text{reshaped}}))) \quad (14)$$

$$\mathbf{x}_{\text{pooled}} = \frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \mathbf{x}_{\text{conv}}[i, j, :] \quad (15)$$

where H' and W' are the dimensions of the feature map after the convolution.

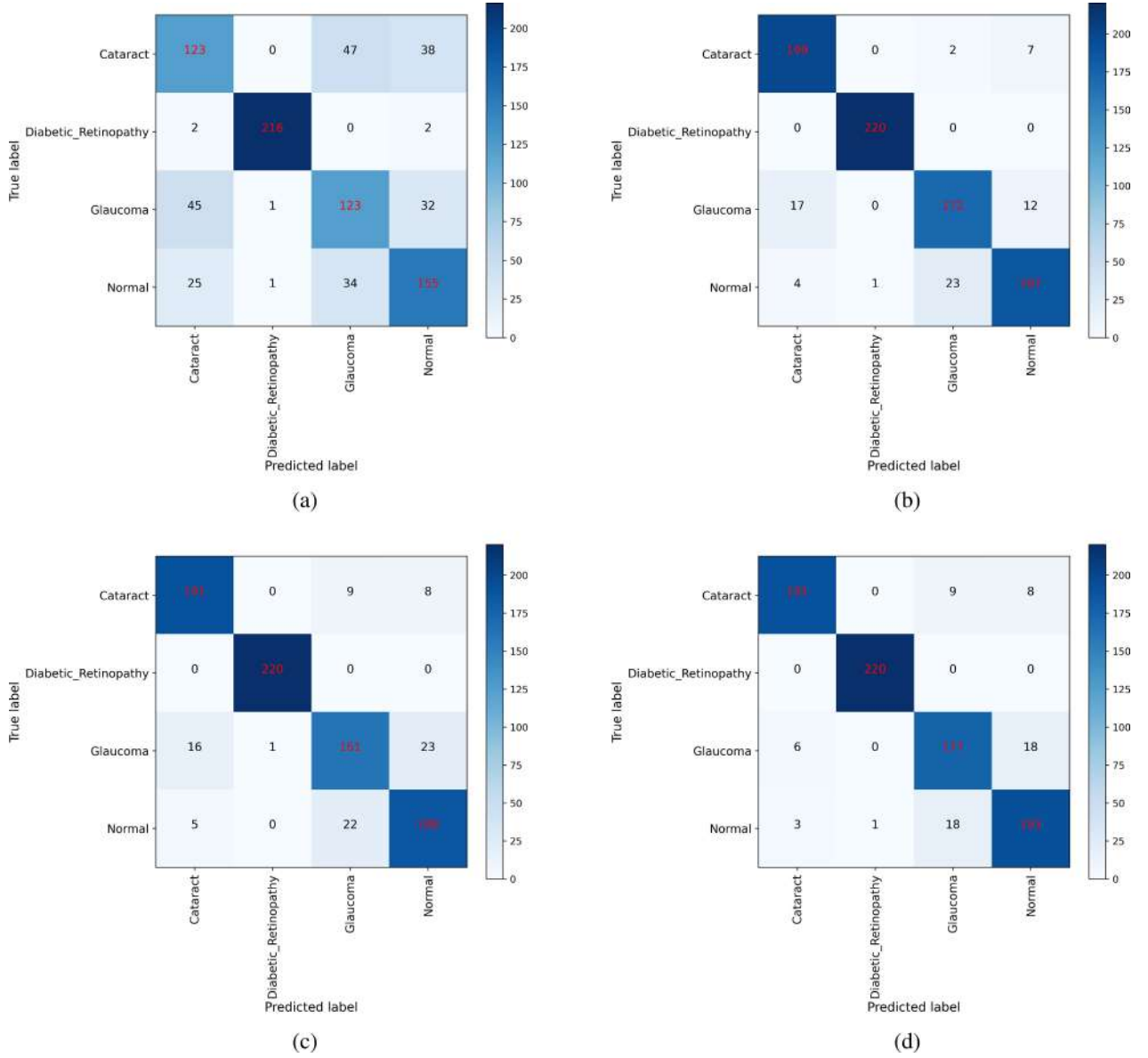


FIGURE 12. Confusion matrix of the proposed models on the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

The refined feature vector is passed through a custom MLP head with a dense layer using ReLU activation and L2 regularization, followed by a dropout layer to prevent overfitting. A final dense layer with a softmax activation generates the probability distribution across the classes. The model is trained by minimizing the categorical cross-entropy loss function

$$\hat{y} = \sigma \left(\text{Dense} \left(\text{MLP}(\mathbf{x}_{\text{pooled}}) \right) \right) \quad (16)$$

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{i,j} \log \hat{y}_{i,j}, \quad (17)$$

In Equation 16, σ denotes the softmax function, which is applied to the output of the final dense layer to produce a predicted probability distribution across classes.

In Equation 17, N denotes the number of samples in the batch and C denotes the number of classes. Furthermore, $y_{i,j}$ and $\hat{y}_{i,j}$ represent the actual ground truth and predicted probabilities for the j -th class of the i -th sample, respectively.

Grad-CAM: For interpretability, we integrate Grad-CAM [25], which highlights the most relevant regions in the input image by computing class score gradient y_c with respect to feature maps A_k :

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k} \quad (18)$$

The Grad-CAM [25] heatmap is computed as

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A_k \right) \quad (19)$$

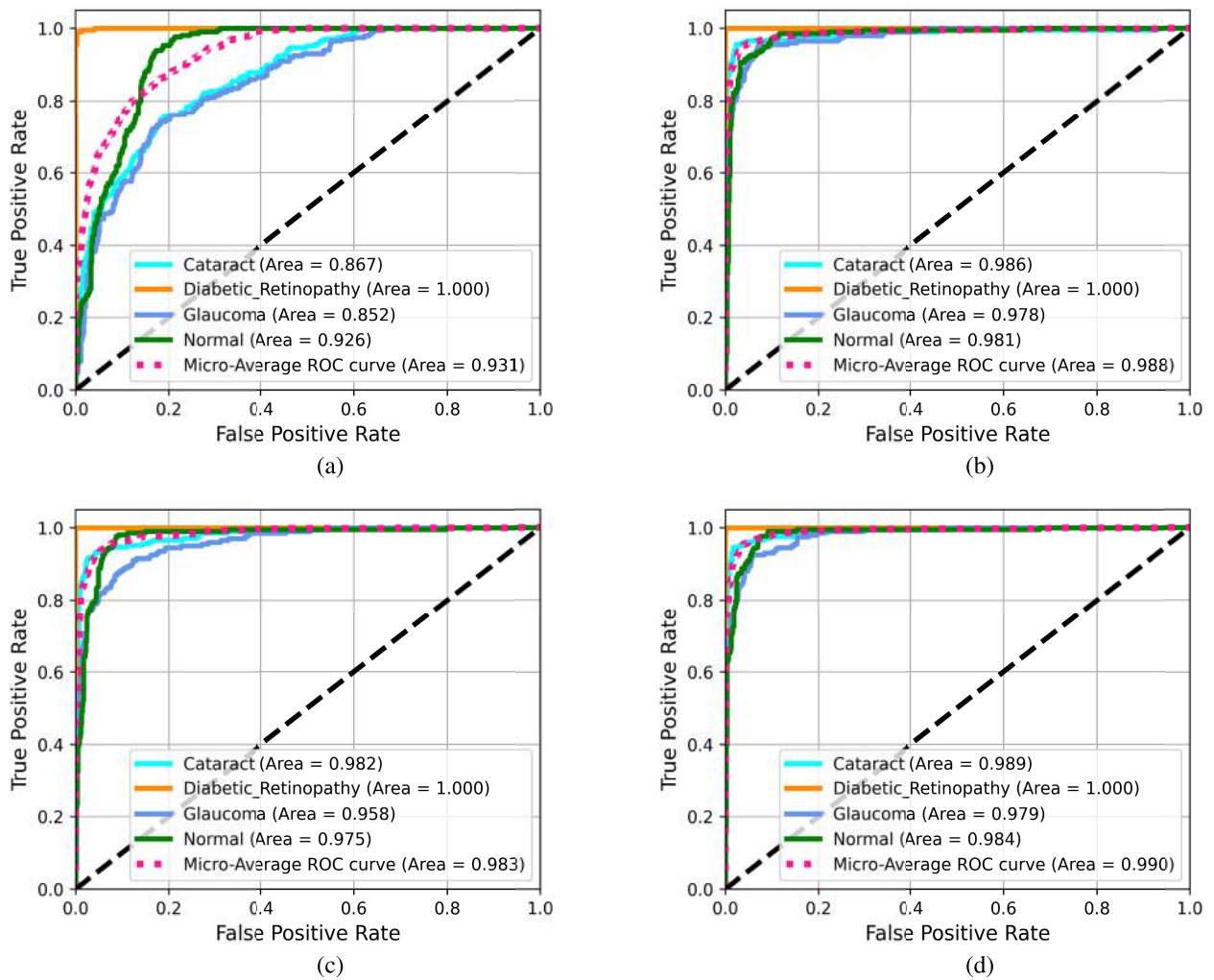


FIGURE 13. ROC curve of the proposed models on the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

The heatmap is superimposed on the input image to visualize regions critical to the model's prediction, enhancing interpretability and fostering clinical trust.

By combining CNNs for local feature extraction with ViT [20] for global attention, the proposed EFFResNet-ViT model achieves superior performance for multiclass tumor classification. Integrating Grad-CAM [25] further enhances the transparency of the model, aligning its decisions with medically relevant features.

IV. EXPERIMENTS

This section describes the experimental setup of the proposed hybrid model, including the dataset, model architecture, training configuration, and evaluation metrics. The configuration parameters of the model are listed in Table 3.

A. DATASET AND PREPROCESSING

The proposed model was evaluated using two publicly available medical image datasets. The first dataset is the BT CE-MRI dataset, available on Kaggle [47]. This dataset was

preorganized into training and testing folders and further classified into four distinct classes: glioma, meningioma, pituitary, and no tumor, as shown in Fig. 5. We further divided the training dataset into 80% for training and 20% for validation, keeping the testing dataset unchanged. A sample of the BT CE-MRI dataset is shown in Fig. 5. Table 1 lists the details of the BT CE-MRI image dataset.

The second dataset is the retinal dataset, available on Kaggle [48]. The dataset consists of cataract, diabetic retinopathy, glaucoma, and normal retinal images; each class has approximately 1000 images. These images were collected from various sources, such as IDRiD, Ocular recognition, and HRF. Fig. 6 shows the sample retinal images dataset. We split the dataset into 60% for training, 20% for validation, and 20% for testing. Table 2 lists the details of the retinal dataset.

We designed our network, performed minimal data pre-processing, reduced the computational costs, and trained our network end-to-end. Preprocessing was conducted as follows. First, all images were resized into 224×224 . Second, we augmented and generated diverse training data

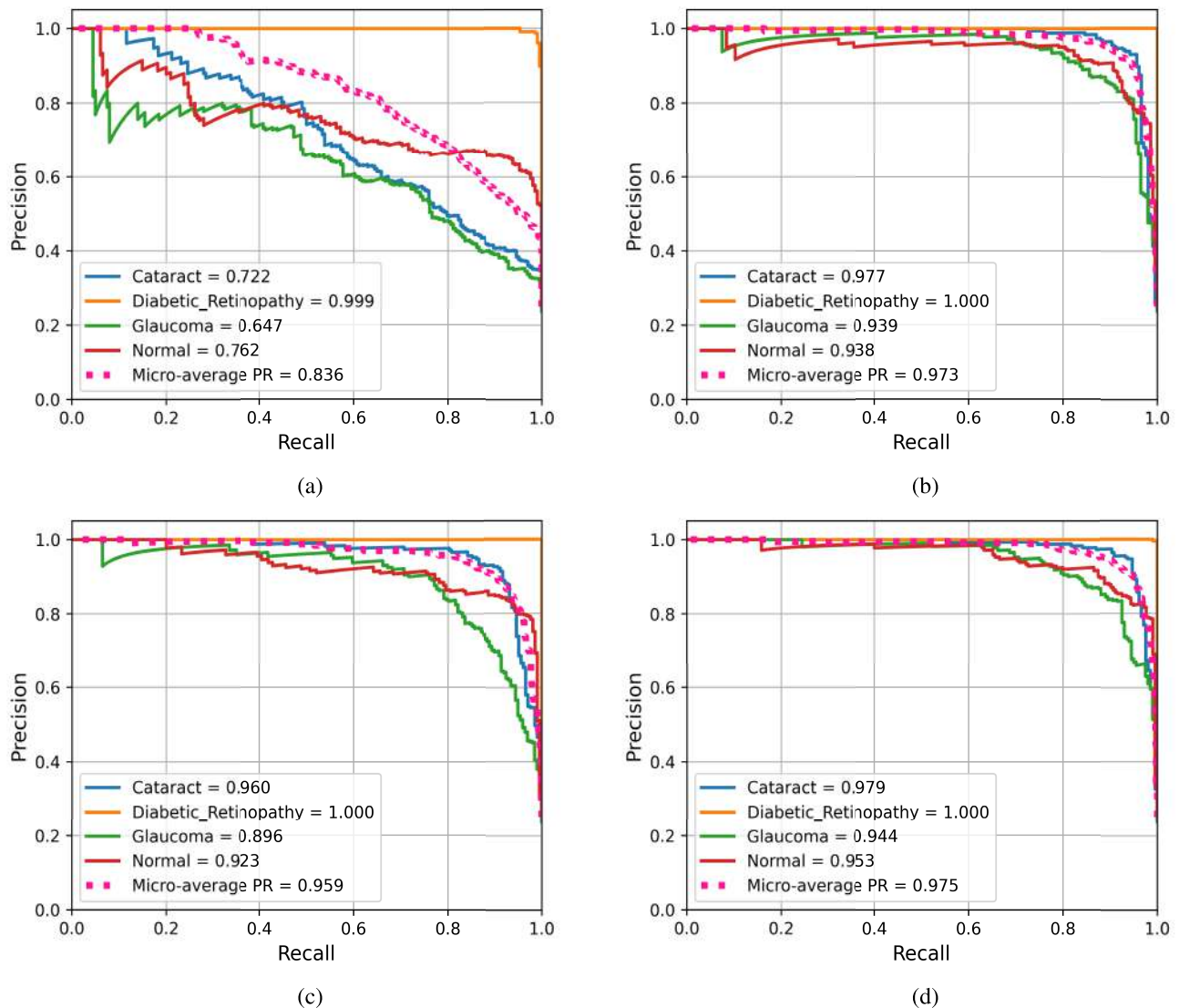


FIGURE 14. Precision-recall curve of the proposed models on the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

by randomly changing the brightness ($0 \sim 0.6$), contrast ($0 \sim 1$), saturation ($0 \sim 0.3$), and hue ($0 \sim 0.1$) on the original input image. Next, we flipped the image vertically and/or horizontally and rotated it -90° or 90° with a probability of 0.5. The final step involves normalizing each RGB pixel value to $[0, 1]$.

Certain parameters were adjusted during the training phase to achieve the best results. In this step, we adjust hyperparameters such as the number of epochs, optimizers, learning rates (LR), patch size, transformer blocks, and MLP dimension. Our classification is based on the softMax activation function. The parameter settings are presented in Table 3.

As shown in Fig. 15, the computational complexity of the medical image classification models varies based on the parameters and inference times. The proposed ViT model, with the fewest parameters ($0.47M$), demonstrates rapid

inference during training ($26.48s$), validation ($7.20s$), and testing ($4.32s$) for the BT CE-MRI dataset, yet exhibits increased computational demands when processing the more complex Retinal dataset with training time of $146.71s$, validation time of $46.68s$, and testing time of $88.66s$. The EfficientNetB0-ViT ($47.27M$ parameters) and ResNet50-ViT ($91.98M$ parameters) models, despite their higher parameter counts, show consistent performance across both datasets, with EfficientNetB0-ViT particularly excelling in computational efficiency on the Retinal dataset, achieving training time of $52.80s$, validation time of $13.74s$, and testing time of $20.78s$. The EffResNet-ViT model achieves a balanced parameter efficiency ($32.75M$) by combining EfficientNet-B0 [23] and ResNet-50 [24] with ViT [20]; however, it consistently requires the highest computational time during training ($495.02s$ for BT CE-MRI, $534.05s$ for Retinal), validation ($183.09s$ for BT CE-MRI, $60.85s$ for

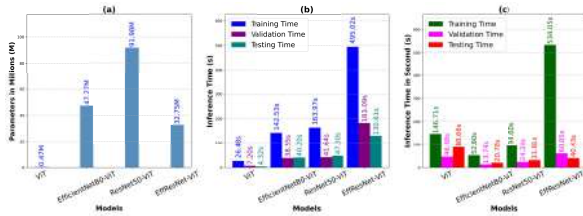


FIGURE 15. Proposed models: (a) Parameters, (b) Inference times for the BT CE-MRI dataset, and (c) Inference times for the Retinal dataset.

Retinal), and testing (130.61s for BT CE-MRI, 40.43s for Retinal) across both datasets as shown in Fig. 15(a) and (b), attributed to the increased complexity introduced by the integrated convolutional feature extractors and transformer architecture.

B. METRICS EVALUATION

Model efficiency was evaluated using various quantitative metrics, including accuracy, precision, recall, F1-score, sensitivity, specificity, Matthews correlation coefficient (MCC), and area under the curve (AUC). For each evaluation criterion, higher scores indicated better performance. These metrics are mathematically defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (21)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (22)$$

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (23)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (24)$$

$$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (25)$$

$$\text{AUC} = \int_0^1 \text{TPR}(t), d(\text{FPR}(t)) \quad (26)$$

where TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively.

V. RESULTS AND DISCUSSION

This section presents a comprehensive analysis of the classification performance of both the BT CE-MRI and retinal datasets. We evaluated various models, including the proposed EffResNet-ViT, and compared them with baseline models such as ViT, EfficientNetB0-ViT, and ResNet50-ViT. Additionally, we present class-wise performance assessments, confusion matrices, ROC curves, precision-recall curves, Grad-CAM [25] visualizations, and t-SNE [26] feature distributions to demonstrate the capabilities of the models further.

A. CLASSIFICATION RESULTS FOR THE BT-CE MRI DATASET

This section presents a detailed analysis of the performance of ViT, EfficientNetB0-ViT, ResNet50-ViT, and the proposed EffResNet-ViT on the multiclass BT CE-MRI dataset. The evaluation is based on metrics mentioned in the Equations 20–26. Visual representations, including confusion matrices, class-wise performance, ROC, and PR-curve, are also presented.

ViT: This model showed average training performance on the BT CE-MRI dataset, as shown in Table 4. The model achieved a training accuracy of 87.68% and a training loss of 0.3279. However, despite its promising training performance, the model struggled to generalize to the validation set, achieving a validation accuracy of 80.71% and a validation loss of 0.5295. This discrepancy suggests potential overfitting because the validation performance was lower than the training performance.

On the test dataset, the ViT model achieved a test accuracy of 80.09%, which, although reasonable, reflects a relatively lower performance compared with the other models, as shown in Table 5. The precision, recall, and F1 scores were 80.31% and 80.09%, and 79.49%, respectively, indicating a balanced performance across the tumor types. The model sensitivity and specificity were 80.09% and 93.26%, respectively, indicating that it was particularly strong at identifying negative tumor cases. The MCC and AUC values of 73.53% and 95.49%, respectively, demonstrate a good overall model performance; however, they were not as precise as the subsequent models.

The confusion matrix shown in Fig. 8(a) further highlights some misclassifications among specific tumor classes, particularly glioma, meningioma, and no-tumor classes. For instance, of the 300 glioma samples, only 191 were correctly identified, with 94 misclassified as meningioma and 13 as no-tumor. Similar patterns were observed for meningioma; of 306 meningioma samples, only 183 were correctly classified, with 17 as misclassified as glioma, 92 as pituitary, and 14 as no-tumor classes. The same trend was observed in normal class, as shown in Fig. 8(a). These observations are further reflected in the class-wise performance metrics shown in Fig. 7(a). Although the ViT model demonstrated reasonable overall accuracy, its classification precision requires improvement, specifically for differentiating between specific tumor types. The MCC (73.53%) and AUC (95.49%) values indicate good overall performance; however, the model was less precise compared to the subsequent models in the study.

The ROC curves, which compare TPR (sensitivity) to FPR, clearly represent the model performance. For the ViT model, as shown in Fig. 9(a), the pituitary and non-tumor classes achieved relatively high ROC values of 0.988 and 0.989, respectively. However, the glioma and meningioma classes exhibited lower ROC values of 0.960 and 0.873, respectively, resulting in a micro-average ROC value of 0.961.

The PR curve for the ViT model, shown in Fig. 10(a), shows similar trends, with PR values of 0.881 for glioma, 0.692 for meningioma, 0.972 for pituitary, and 0.950 for no-tumor, resulting in a micro-average PR value of 0.897. This performance indicates that while the ViT model excels in recognizing pituitary and no-tumor classes, it struggles with glioma and meningioma. The relatively poor performance for glioma and meningioma can be attributed to the limited ability of the ViT model to capture finer local features, which is essential for accurately distinguishing these tumor types.

EfficientNetB0-ViT: This model demonstrated remarkable performance, significantly outperforming the ViT model, as shown in Table 4. The training accuracy reached 99.82%, with a very low training loss of 0.0301, suggesting that the model effectively learned the training data. The validation accuracy of 96.14% with a validation loss of 0.1961 indicated that the model generalized well to unseen data, making it a robust classifier for BT.

Table 5 presents the test performance of the EfficientNetB0-ViT model, which achieved an impressive test accuracy of 98.32%, which is significantly higher than that of the ViT model. The precision, recall, and F1-score were excellent, reaching 98.32%, 98.32%, and 98.30%, respectively, confirming the model's highly reliable classification of tumor types. The sensitivity and specificity of 98.32% and 99.45%, respectively, further highlighted the ability of the model to accurately differentiate between tumor and non-tumor cases. Moreover, the MCC and AUC of 97.75% and 99.95%, respectively, underscored the strong discriminative capability of the model.

The confusion matrix and class-wise performance are shown in Fig. 8(b) and Fig. 7(b), demonstrating that the EfficientNetB0-ViT model maintained high precision across all tumor classes.

The ROC curve of the EfficientNetB0-ViT model, shown in Fig. 9(b), highlights its exceptional performance across all classes. The glioma and meningioma classes achieved a near-perfect ROC value of 0.999. In contrast, the pituitary and no-tumor classes attained a perfect ROC value of 1.000, resulting in a microaverage ROC value of 0.999.

The PR curve of the EfficientNetB0-ViT model, shown in Fig. 10(b), mirrors this outstanding performance, with PR values of 0.997 for glioma, 0.998 for meningioma, 0.999 for pituitary, and 1.000 for no-tumor. The micro-average PR value is 0.998, demonstrating the superior ability of the model to differentiate between tumor types.

This significant performance improvement can be attributed to the enhanced local feature extraction capabilities of EfficientNet-B0, which effectively captures fine-grained details in tumor regions, combined with the global context modeling strengths of ViT.

ResNet50-ViT: This model performed well, although slightly behind the EfficientNetB0-ViT model, as shown in Table 4. The training accuracy of 98.31% and a training

loss of 0.1419 indicate that the model learned from the dataset with a relatively low error. The validation accuracy of 92.11% with a validation loss of 0.3359 demonstrates good generalization to unseen data, although it is not as strong as that of the EfficientNetB0-ViT model.

Table 5 presents the test performance of the ResNet50-ViT model, which achieved a test accuracy of 95.58%, with precision, recall, and F1-score values of 95.65%, 95.58%, and 95.55%, respectively. The sensitivity and specificity of 95.58% and 98.53%, respectively, were also strong but not as high as those of the EfficientNetB0-ViT model. The MCC and AUC of 94.10% and 99.73%, respectively, still indicated good performance.

The confusion matrix and class-wise performance are shown in Fig. 8(c) and Fig. 7(c), demonstrating that the ResNet50-ViT model is still a strong performer; however, it falls slightly behind the EfficientNetB0-ViT model because of the less efficient local feature extraction of ResNet-50 compared to EfficientNet-B0, affecting the overall generalization of the model.

The ROC curve of the ResNet50-ViT model, shown in Fig. 9(c), also demonstrates strong performance, with ROC values of 0.998 for glioma and pituitary, 0.993 for meningioma, and 0.999 for no-tumor, resulting in a micro-average ROC value of 0.997.

The PR curve shown in Fig. 10(c) shows similar strong results, with PR values of 0.992 for glioma, 0.982 for meningioma, 0.997 for pituitary, and 0.998 for no-tumor, leading to a micro-average PR value of 0.992. The strong performance of this model is attributed to the combination of ResNet-50 robust local feature extraction and ViT global contextual attention mechanism, which enables accurate classification even for complex tumor types, as compared with the ViT model shown in Fig. 10(a).

EffResNet-ViT: EffResNet-ViT, which combines the strengths of EfficientNet-B0 and ResNet-50 as the backbone of the ViT model, demonstrated the best performance among all the models, as shown in Table 4. The training accuracy of 100% with a very low training loss of 0.0097 indicates that the model learned the dataset's features exceptionally well with minimal loss. The validation accuracy of 99.21% and validation loss of 0.0467 suggest outstanding generalization to unseen data, making it the most reliable model for training and validation.

Table 5 summarizes the test performance of the proposed EffResNet-ViT model, which achieved a test accuracy of 99.31%, the highest among all the proposed models, with excellent precision, recall, and F1-score values of 99.26%, 99.25%, and 99.26%, respectively. The sensitivity and specificity of 99.25% and 99.78%, respectively, further highlighted the accuracy of the model in distinguishing between tumor and no-tumor cases, with an MCC and AUC of 99.08% and 99.91%, respectively. These results reveal that the EffResNet-ViT model exhibited exceptional performance across all metrics compared to the other proposed models, as shown in Table 5.

The confusion matrix in Fig. 8(d) and class-wise performance in Fig. 7(d) confirmed that the model achieved near-perfect precision across all tumor types.

Fig. 9(d) shows that the proposed EffResNet-ViT model achieved the best ROC values of 1.000 for glioma and pituitary and near-perfect values of 0.996 and 0.997 for meningioma and no-tumor, respectively, resulting in a micro-average ROC value of 0.999.

The PR curve of the EffResNet-ViT model, shown in Fig. 10(d), demonstrates exceptional performance with PR values of 1.000 for glioma, 0.996 for meningioma, 1.000 for pituitary, and 0.997 for no-tumor, resulting in a micro-average PR value of 0.998.

The proposed EffResNet-ViT model consistently outperformed the ViT, EfficientNetB0-ViT, and ResNet50-ViT models in terms of training, validation, and testing. Its hybrid approach, merging the strengths of CNNs and ViT, enabled it to achieve the highest accuracy and best generalization to unseen data. This confirms the effectiveness of the EffResNet-ViT model for the multiclass BT CE-MRI dataset classification.

B. CLASSIFICATION RESULTS FOR THE RETINAL DATASET

This section presents a detailed evaluation of the performance of the proposed EffResNet-ViT model alongside three baseline models on a multiclass retinal classification task. The evaluation is based on metrics mentioned in the Equations 20–26. Visual representations are also presented.

ViT: The model demonstrated moderate performance during training and validation, as detailed in Table 4. It achieved training and validation accuracies of 90.54% and 73.92%, respectively, with corresponding losses of 0.2590 and 0.8262, respectively, indicating significant overfitting and poor generalization to the validation set.

The test performance, as presented in Table 5, revealed an accuracy of 73.10% with precision, recall, and F1-score values of 73.13%, 73.10%, and 73.08%, respectively, and a specificity of 91.08%. The MCC and AUC scores were 64.14% and 91.30%, respectively, indicating a moderate performance compared to the other proposed models.

However, the confusion matrix presented in Fig. 12(a) reveals significant misclassifications, particularly among the cataract, glaucoma, and normal classes. For example, of the 208 cataract samples, only 123 were correctly identified, with 47 misclassified as glaucoma and 38 as normal. Similar patterns were observed for the glaucoma and normal classes.

The ROC curve of the ViT model, shown in Fig. 13(a), demonstrates varied performance across the retinal dataset classes. The model achieved a perfect ROC value of 1.000 for diabetic retinopathy, whereas the cataract, glaucoma, and normal classes exhibited moderate ROC values of 0.867, 0.852, and 0.926, respectively. The micro-average ROC value was 0.931, indicating good overall performance in distinguishing between different retinal conditions, with extreme discrimination between diabetic retinopathy and normal classes.

The PR curve for the ViT model, shown in Fig. 14(a), demonstrates weaker performance compared with the ROC curve. Although diabetic retinopathy achieves a near-perfect PR value of 0.999, the model performs relatively poorly for the other classes, with cataract, glaucoma, and normal classes yielding PR values of 0.722, 0.647, and 0.762, respectively. The micro-average PR value is 0.836, suggesting that the ability of the model to accurately classify retinal conditions was less robust, particularly for the glaucoma and cataract classes. This is likely because of the model's difficulty in distinguishing between these less prevalent conditions, resulting in a higher rate of FP and FN, which adversely affects precision and recall.

EfficientNetB0-ViT: The model showed substantial improvements over ViT, achieving training and validation accuracies of 99.96% and 92.74%, respectively, with a minimal gap between training and validation losses, which are 0.0720 and 0.3402, respectively, suggesting strong generalization.

During testing, it achieved an accuracy, a precision, and a recall of 92.18%, 92.16%, and 92.18%, respectively, with an MCC and AUC of 89.59% and 98.62%, respectively, reflecting a robust and balanced performance. The confusion matrix in Fig. 12(b) and class-wise performance in Fig. 11(b) demonstrates that the EfficientNetB0-ViT model maintained high precision across all tumor classes.

The ROC curve of the EfficientNetB0-ViT model, shown in Fig. 13(b), reveals strong performance, with ROC values of 0.986 for cataract, 1.000 for diabetic retinopathy, 0.978 for glaucoma, and 0.981 for normal, resulting in a micro-average ROC value of 0.988.

The PR curve of the EfficientNetB0-ViT model, shown in Fig. 14(b), demonstrates improved performance compared with ViT, with PR values of 0.977 for cataract, 1.000 for diabetic retinopathy, 0.939 for glaucoma, and 0.938 for normal. The micro-average PR value is 0.973, highlighting the superior ability of the model to distinguish between retinal conditions and achieve a balanced trade-off between precision and recall.

ResNet50-ViT: The model displayed strong training performance with an accuracy of 99.88% and validation accuracy of 91.25%, with slightly higher validation loss of 0.4502 compared to EfficientNetB0-ViT shown in Table 4.

The test performance is presented in Table 5. It achieved an accuracy, precision, recall, F1-score, specificity, MCC, and AUC of 90.05%, 89.99%, 90.05%, 90.00%, 96.70%, 86.73%, and 97.92%, respectively.

The confusion matrix in Fig. 12(c) and class-wise performance in Fig. 11(c) reveal that the ResNet50-ViT model maintained high precision across all retinal conditions, although it was slightly lower than that of EfficientNetB0-ViT and higher than that of the ViT model.

The ROC curve shown in Fig. 13(c) reveals good performance compared to the ViT model but slightly lags behind the EfficientNetB0-ViT, with class-specific ROC values of 0.982 for cataract, 1.000 for diabetic retinopathy, 0.958 for

glaucoma, and 0.975 for normal, leading to a micro-average ROC value of 0.983.

The PR curve, shown in Fig. 14(c), indicates good performance when compared to the ViT model but does not surpass the EfficientNetB0-ViT model, with PR values of 0.960 for cataract, 1.000 for diabetic retinopathy, 0.896 for glaucoma, and 0.923 for normal, resulting in a micro-average PR value of 0.959.

The slightly lower performance compared to the EfficientNetB0-ViT model can be attributed to the superior ability of EfficientNetB0 to capture fine-grained details, which leads to better generalization and classification accuracy. However, the ResNet50-ViT model outperforms the ViT model because of the inclusion of ResNet50's convolutional layers, which are better at capturing local spatial features than ViT's global attention mechanism.

EffResNet-ViT: The model outperformed all baseline models for the retinal dataset, achieving training and validation accuracies of 99.96% and 92.29%, respectively, with the lowest training and validation losses of 0.0363 and 0.3284, respectively, as shown in Table 4.

Table 5 summarizes the test performance. They achieved an accuracy, precision, recall, specificity, MCC, and AUC of 92.54%, 92.60%, 92.54%, 97.52%, 90.05%, and 98.82%, respectively, highlighting their exceptional generalization ability compared to other proposed models.

The confusion matrix shown in Fig. 12(d) shows the fewest misclassifications compared to the other models, demonstrating the superior classification ability of EffResNet-ViT.

The class-wise performance for all metrics further revealed that the proposed model outperformed the other models across all metrics. The class-wise performance of the proposed EffResNet-ViT is shown in Fig 7(d).

The ROC curve of the proposed EffResNet-ViT model on the retinal dataset, shown in Fig. 13(d), highlights the excellent ability of the model to distinguish different retinal conditions. The model achieves high ROC values for all classes, with cataract of 0.989, diabetic retinopathy of 1.000, glaucoma of 0.979, and normal of 0.984, indicating strong classification performance for each condition. The microaverage ROC value of 0.990 further reinforces the robust overall performance of the model, demonstrating its capability to correctly discriminate between retinal conditions with minimal errors.

The PR curve of the EffResNet-ViT model, shown in Fig. 14(d), also indicates high performance, particularly in handling the imbalanced nature of the dataset. The model achieves a PR value of 1.000 for diabetic retinopathy and relatively high values for the other classes: 0.979 for cataract, 0.944 for glaucoma, and 0.953 for normal. These results indicate that the EffResNet-ViT model effectively balances PR across all retinal conditions, with a micro-average PR value of 0.975, suggesting an excellent overall classification capability. This demonstrates that the model is accurate and reliable for identifying the presence of retinal diseases with high confidence across all classes.

C. EXPLAINABILITY RESULTS ANALYSIS

This section presents a comprehensive analysis of the explainability results using Grad-CAM [25] for the ViT, EfficientNetB0-ViT, ResNet50-ViT, and proposed EffResNet-ViT model. Grad-CAM [25] is a powerful tool for visualizing regions in input images that contribute most significantly to the model's predictions, thereby enhancing the transparency and interpretability in medical image classification tasks. The Grad-CAM [25] results for BT CE-MRI and retinal datasets are presented in Fig. 16.

Grad-CAM [25] heatmaps for the ViT model, depicted in Fig. 16(b), exhibit scattered focus across both datasets. For the BT CE-MRI dataset, the model frequently highlights non-relevant regions outside the tumor area, thereby reducing its diagnostic reliability. Similarly, in the retinal dataset, the heatmaps demonstrate diffuse focus patterns, with the model often failing to consistently emphasize pathological regions. The lack of precise localization diminishes the interpretability and clinical applicability of the ViT model.

The EfficientNetB0-ViT model, as illustrated in Fig. 16(c), shows improved localization compared to the ViT model. For the BT CE-MRI dataset, the Grad-CAM [25] heatmaps focus better on tumor boundaries; however, occasional spillover into nonrelevant areas persists. In the retinal dataset, the model demonstrates enhanced attention to pathological regions, thereby capturing critical features more effectively. Despite this improvement, there are instances where irrelevant regions are highlighted, indicating the need for further optimization.

The Grad-CAM [25] results for the ResNet50-ViT model, shown in Fig. 16(d), indicate higher precision in feature localization. On the BT CE-MRI dataset, the model accurately highlights tumor regions while minimizing attention to irrelevant areas, thereby aligning with clinical expectations. The heatmaps are more specific and consistent for the retinal dataset, effectively identifying the pathological regions. Nonetheless, the model occasionally focuses on irrelevant regions in complex or ambiguous cases, which may affect interpretability in specific scenarios.

The Grad-CAM [25] visualizations for the proposed EffResNet-ViT model, presented in Fig. 16(e), exhibit the most refined and precise focus on both datasets. On the BT CE-MRI dataset, the model consistently localizes tumor regions with high fidelity, as evidenced by heatmaps that closely align with the actual tumor locations. Similarly, the model reliably highlights pathological regions in the retinal dataset, capturing subtle features critical to each class. This enhanced localization significantly improves the interpretability and clinical reliability of the EffResNet-ViT model.

The integration of the CNN and transformer features in the EffResNet-ViT model enables it to address the limitations observed in other models. Its Grad-CAM [25] heatmaps provide clear and clinically relevant visualizations, thereby enhancing the transparency and trust in its diagnostic

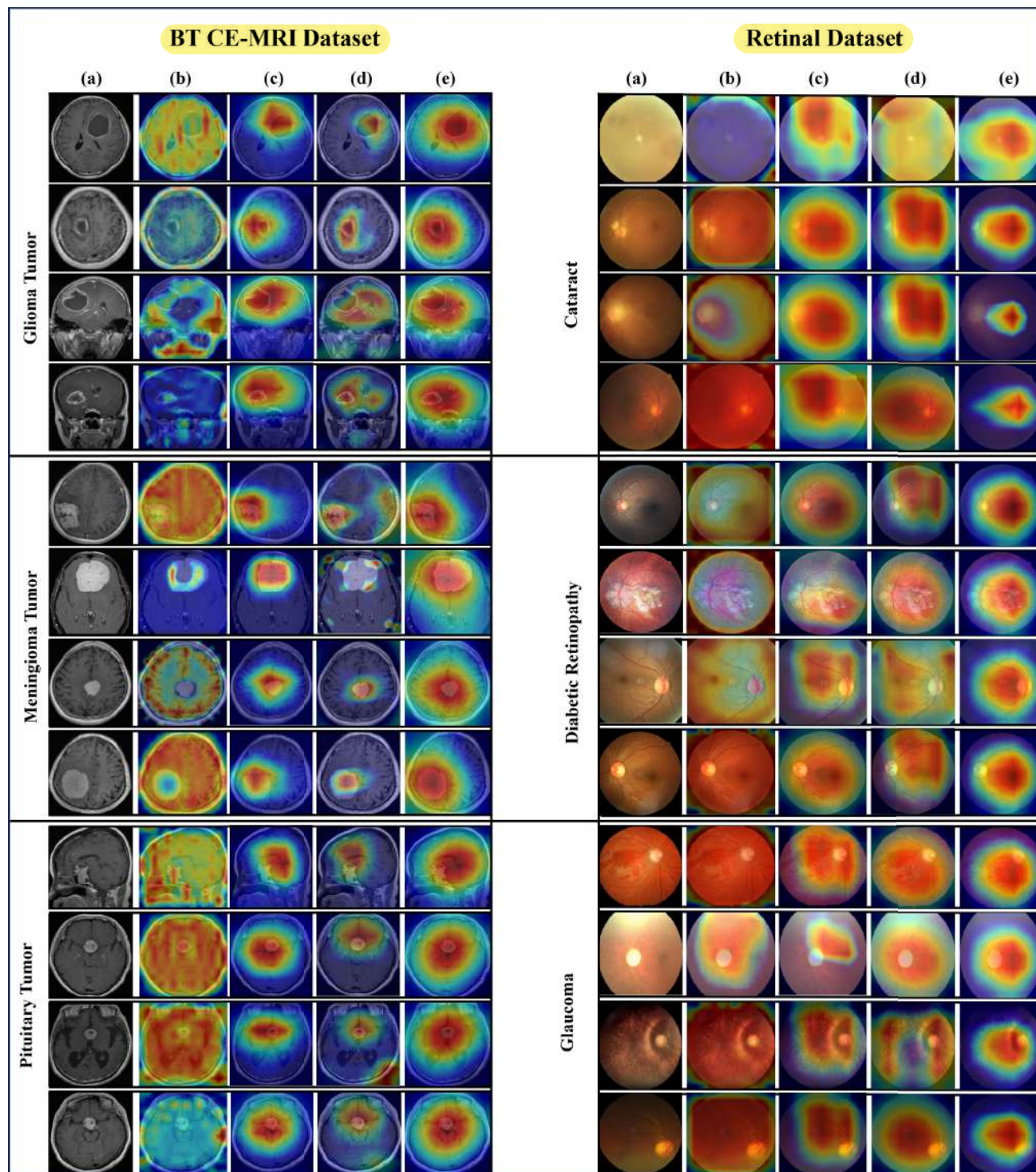


FIGURE 16. The Grad-CAM results of the proposed models on the BT CE-MRI and retinal datasets: (a) input classes of both the MRI and retinal datasets, (b) ViT, (c) EfficientNetB0-ViT, (d) ResNet50-ViT, and (e) EffResNet-ViT.

capabilities. These findings highlight the strengths and weaknesses of the evaluated models, with the EffResNet-ViT model demonstrating superior localization and interpretability, closely aligning with the diagnostic expectations for medical image classification.

D. FEATURE MAPPING WITH T-SNE

t-SNE [26] is an unsupervised, nonlinear dimensionality reduction technique that is widely used to visualize high-dimensional data in a lower-dimensional space, typically in 2D or 3D. It preserves the pairwise similarities

between data points, allowing for easier interpretation of clustering patterns. This study applies t-SNE [26] to examine the feature representations of different models for the BT CE-MRI and retinal datasets. The visualizations provide insight into how well the models separate distinct classes, such as glioma, meningioma, pituitary, and no tumor, in the BT CE-MRI dataset and various pathological conditions in the retinal dataset.

1) BT CE-MRI DATASET

The t-SNE [26] results for the BT CE-MRI dataset, shown in Fig. 17(a), corresponding to the ViT model, showed considerable overlap between the feature clusters of different tumor classes, particularly among glioma, meningioma, pituitary, and no tumor. The data points are scattered and mixed, indicating that the ViT model struggles to effectively distinguish between these tumor types. Weak feature separation makes it difficult for the model to reliably differentiate between conditions. This overlap implies that the model does not capture sufficient discriminative features, leading to poor classification performance.

Fig. 17(b) shows a t-SNE [26] plot of the EfficientNetB0-ViT model. It shows a further improvement in feature separation compared to the ViT and ResNet50-ViT model results, as shown in Fig. 17(a) and (b). Glioma, meningioma, pituitary, and no tumor clusters are more distinct. However, some misclassified points still exist within glioma, meningioma, pituitary, and no tumor; however, no cluster overlaps are observed. The EfficientNetB0-ViT model effectively learns more discriminative features; however, it struggles to perfectly separate all classes, particularly in complex cases where tumor characteristics are not easily distinguished.

Fig. 17(c) presents the t-SNE [26] plot for the ResNet50-ViT model. Compared to the ViT model, a noticeable improvement in the separation of feature clusters exists. However, some overlap between the meningioma and no tumor classes persists. This indicates that although the ResNet50-ViT model benefits from the added convolutional layers for feature extraction, it struggles to completely distinguish these classes, specifically in cases where the tumor characteristics are not easy to identify.

The t-SNE [26] results for the EffResNet-ViT model, shown in Fig. 17(d), exhibit the best clustering across all tumor classes. The glioma, pituitary, and no-tumor clusters are tightly grouped with no overlap, demonstrating that this model excels at learning robust, discriminative features. Although some misclassification points between the glioma and meningioma clusters occur, it is minimal compared to the other models shown in Fig. 17(a-c), and the overall clustering is much cleaner. This indicates that EffResNet-ViT is the most effective in distinguishing between different tumor types, with feature points tightly grouped within their respective classes and better intra-class compactness.

2) RETINAL DATASET

The t-SNE [26] results of the ViT model for the retinal dataset are shown in Fig 18(a). This reveals a significant

overlap between feature clusters for different pathological conditions. The data points are spread across 2D space with no clear separation, indicating that the ViT model fails to capture the underlying structures of the retinal images. The lack of distinct clustering and ease of blending feature points for different classes suggest that the model cannot reliably differentiate between various conditions.

In Fig. 18(b), the EfficientNetB0-ViT model shows some improvement in clustering compared to the ViT model. The clusters for different pathological conditions are more separated, although some overlap exists, particularly between the glaucoma and normal classes. This indicates that the model benefits from the convolutional backbone, leading to better clustering. However, these features could not be fully separated under the more complex conditions in the retinal dataset.

The t-SNE [26] plot of the ResNet50-ViT model, shown in Fig. 18(c), reveals a noticeable improvement in the separation of feature clusters compared with the ViT model, although it still lags behind the EfficientNetB0-ViT model. However, there is some overlap between the cataract, diabetic retinopathy, and glaucoma classes, with a few misclassified points in the normal class. Despite this, the model shows more substantial feature discrimination than the ViT model; however, it still struggles to distinguish certain classes in more complex cases.

Finally, the t-SNE [26] results for the proposed EffResNet-ViT model, shown in Fig. 18(d), reveal better separation of feature clusters compared to the other models results shown in Fig. 18(a-c). The classes are well separated, with minimal overlap between the glaucoma and normal classes, and well-clustered cataract and diabetic retinopathy classes, suggesting that the model has effectively learned distinct features for each condition. The clustering of feature points indicates that EffResNet-ViT successfully differentiates complex patterns in retinal images. This demonstrates the superior feature learning performance and the ability of the model to distinguish between pathological conditions.

E. FEATURES ANALYSIS

For the feature analysis of the BT CE-MRI and retinal datasets, we investigated the extraction of features from various layers across ViT, EfficientNetB0-ViT, ResNet50-ViT, and the proposed EffResNet-ViT. Feature visualizations are shown in Figs. 19 and 20 for the BT CE-MRI and retinal datasets, respectively.

In the ViT model, feature extraction begins with the `Initial_layer`, which captures low-level details, such as textures and edges, essential for identifying fundamental tumor or retinal features. Finally, a `Conv2D` layer with `BN_layer` and `ReLU_layer` is applied after the transformer encoder to further refine the features by focusing on critical local patterns. This enhanced feature extraction process is shown in Fig. 19(a) for the BT CE-MRI dataset and Fig. 20(a) for the retinal dataset.

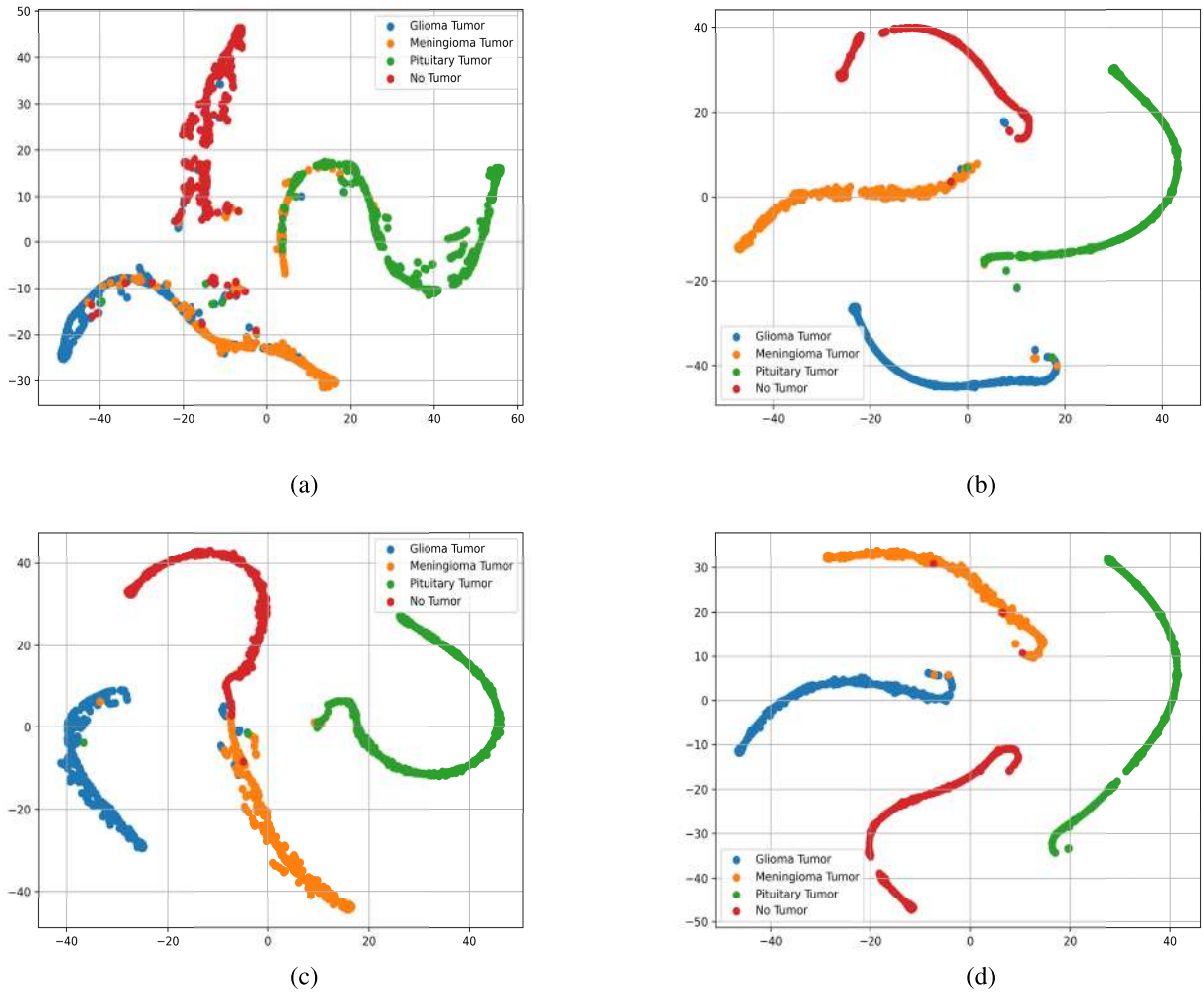


FIGURE 17. t-SNE features of the proposed models on the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

For the EfficientNetB0-ViT model, the features are extracted from various blocks within the EfficientNetB0 architecture. Block_2 captures low-level details, whereas Block_4 and Block_6 represent increasingly abstract and complex patterns. Block_7 captures high-level features crucial for identifying detailed tumor structures or advanced retinal conditions. After passing through the transformer encoder, Conv2D layers are applied to further refine the features by focusing on specific spatial patterns, as shown in Figs. 19(b) and 20(b) for both datasets.

In the ResNet50-ViT model, features are extracted from multiple stages of ResNet50. Block_2 captures early-stage low-level features, whereas Block_3 and Block_4 capture more complex patterns. Block_5 captures high-level spatial features essential for distinguishing between advanced tumor types or complex retinal anomalies. After passing through the transformer encoder, the features are refined through a Conv2D layer, enhancing the model's focus on relevant spatial details, as depicted in Figs 19(c) and 20(c) for the BT CE-MRI and retinal datasets, respectively.

For the proposed EffResNet-ViT model, feature extraction begins with high-level features from Block_7 of EfficientNetB0, which are important for understanding the spatial structure of tumors or retinal features. The features from Block_5 of ResNet50 are then visualized, representing refined, complex patterns. These two sets of features are concatenated, enabling the model to capture complementary information from both architectures. Finally, a Conv2D layer is applied after the transformer encoder refines the features by focusing on the critical local patterns. This enhanced feature extraction process is shown in Fig 19(d) for the BT CE-MRI dataset and Fig 20(d) for the retinal dataset.

By combining CNN-based feature extraction from EfficientNet-B0 and ResNet-50 with global context modeling provided by the transformer encoder, the proposed EffResNet-ViT model leverages both local and global patterns. This multiscale approach significantly enhances the ability of the model to classify complex tumor and retinal conditions, leading to improved model performance,

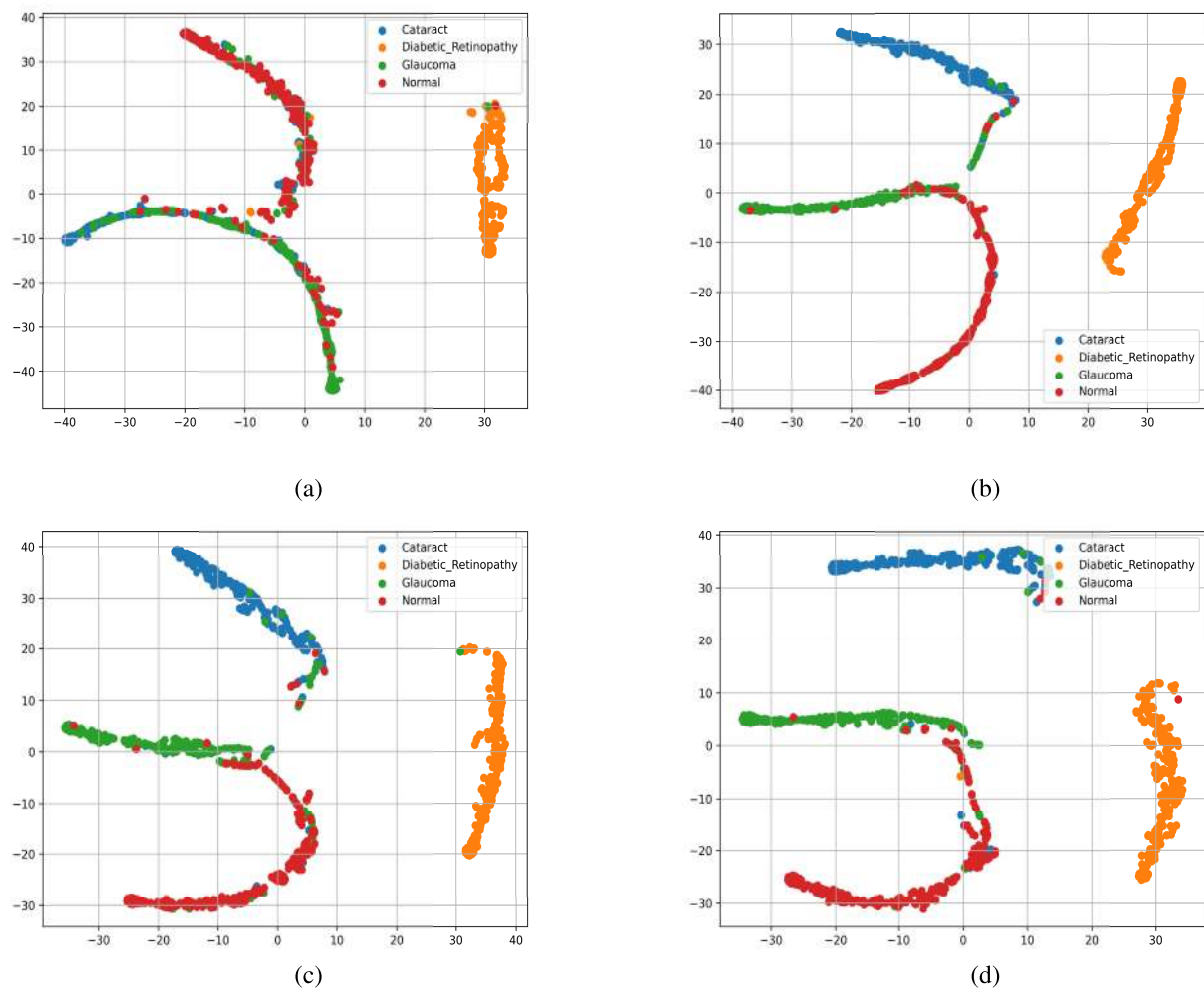


FIGURE 18. t-SNE features of the proposed models on the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

as demonstrated in Figs 19 and 20 for the BT CE-MRI and retinal datasets, respectively.

F. ABLATION STUDY

We demonstrate the effectiveness of the hybrid architecture combining EfficientNet-B0 (efB0), ResNet-50 (R50), and ViT through an ablation study. This study systematically evaluates the impact of different configurations by fine-tuning the individual components—efB0, R50, and ViT—along with their combinations. The core of this approach involves concatenating the features extracted from efB0 and R50 and then feeding them into a fine-tuned ViT to evaluate how this fusion of local and global feature extraction capabilities improves performance.

The ablation study for EFFResNet-ViT investigates the contribution of each architectural component to the overall model performance in medical image classification tasks. Various model setups, including standalone ViT, efB0-ViT, R50-ViT, and the feature concatenation of efB0 and R50, are tested, followed by fine-tuning in ViT. These configurations are then evaluated on standard performance metrics,

including accuracy (Acc), precision (Prec), recall (Rec), F1-score (F1), specificity (Spec), MCC, and Area AUC.

Table 6 presents the results of the ablation study on the BT CE-MRI dataset, showing the influence of different model configurations on classification performance. Standalone models, such as ViT, efB0, and R50, exhibit varied performance, with the ViT model achieving an accuracy of 80.09%, efB0-ViT reaching 98.32%, and R50-ViT performing at 95.58%. However, the most substantial performance gain is observed with the proposed EFFResNet-ViT model, which combines the features of both efB0 and R50 before being fine-tuned by ViT. This model achieves an Acc of 99.31%, along with significant improvements in all other metrics, including Pre, Rec, F1, Spec, MCC, and AUC shown in Table 7.

Table 7 presents the results of the ablation study for the retinal dataset, showing similar trends. The standalone ViT achieves an accuracy of 73.10%, efB0-ViT reaches 92.18%, and R50-ViT performs at 90.05%. When combining ViT with efB0 and R50, the proposed EFFResNet-ViT model outperforms all other configurations, achieving an Acc of

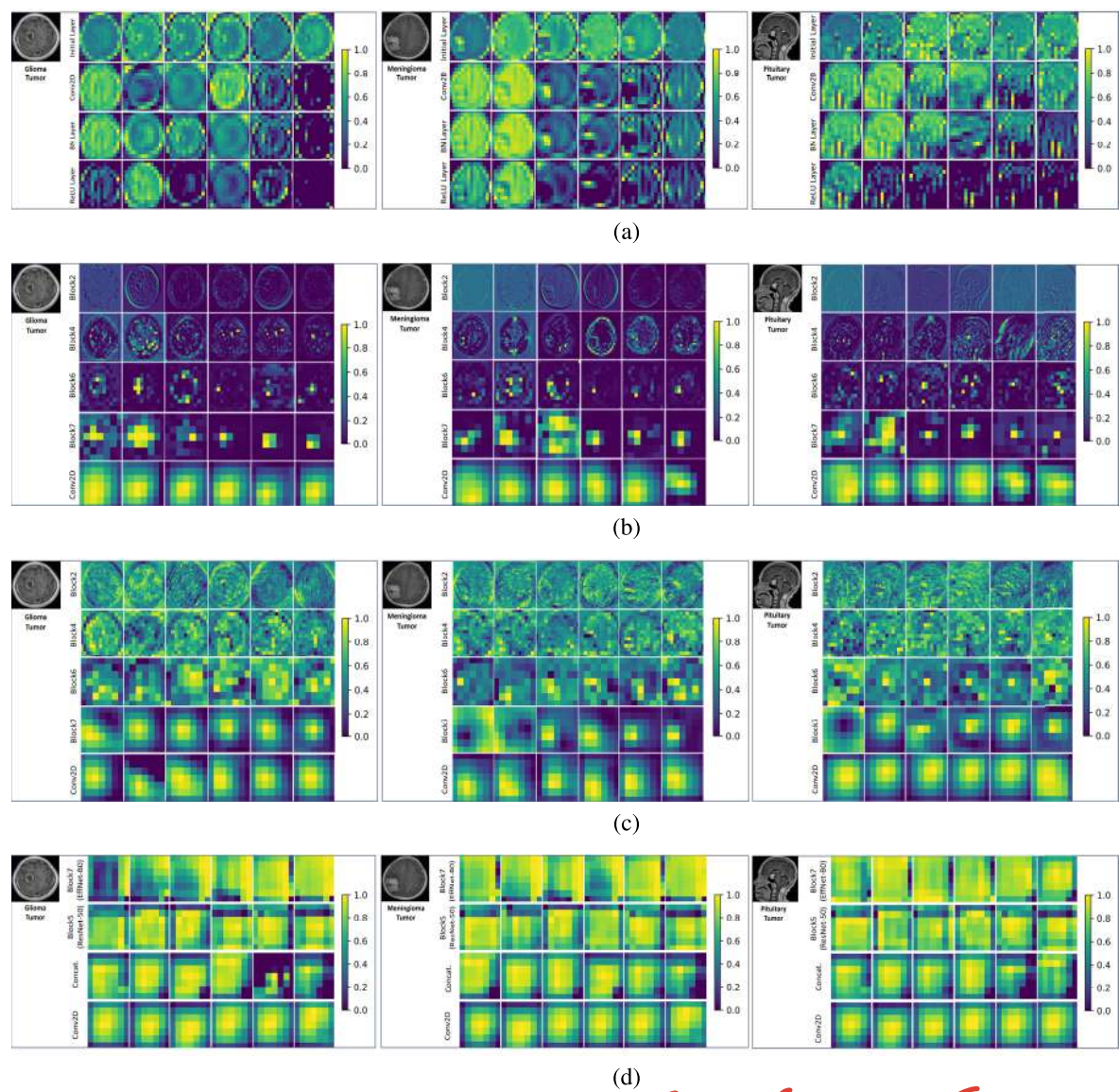


FIGURE 19. Features of the proposed models on the BT CE-MRI dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

TABLE 6. Ablation study performance comparison of proposed models on the BT CE-MRI dataset. The best performance is highlighted in bold.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Spec (%)	MCC (%)	AUC (%)
ViT w/o efB0 & R50	80.09	80.31	80.09	79.49	93.26	73.53	95.49
efB0 + ViT (no R50)	98.32	98.32	98.32	98.30	99.45	97.75	99.95
R50 + ViT (no efB0)	95.58	95.65	95.58	95.55	98.53	94.10	99.73
Proposed EffResNet-ViT (efB0 + R50 + ViT)	99.31	99.26	99.25	99.26	99.78	99.08	99.91

92.54%, along with significant improvements in all other performance metrics, such as Pre, Rec, F1, Spec, MCC, and AUC shown in Table 7.

The results of these ablation studies validate the synergistic effect of combining efB0 and R50 features for fine-tuning in ViT. The fusion of CNN-based local feature extraction

(from efB0 and R50) with the global context modeling capabilities of ViT significantly enhances the classification performance of the model. This hybrid approach improves feature extraction and ensures efficient learning, making EFFResNet-ViT a highly effective model for medical image classification tasks.

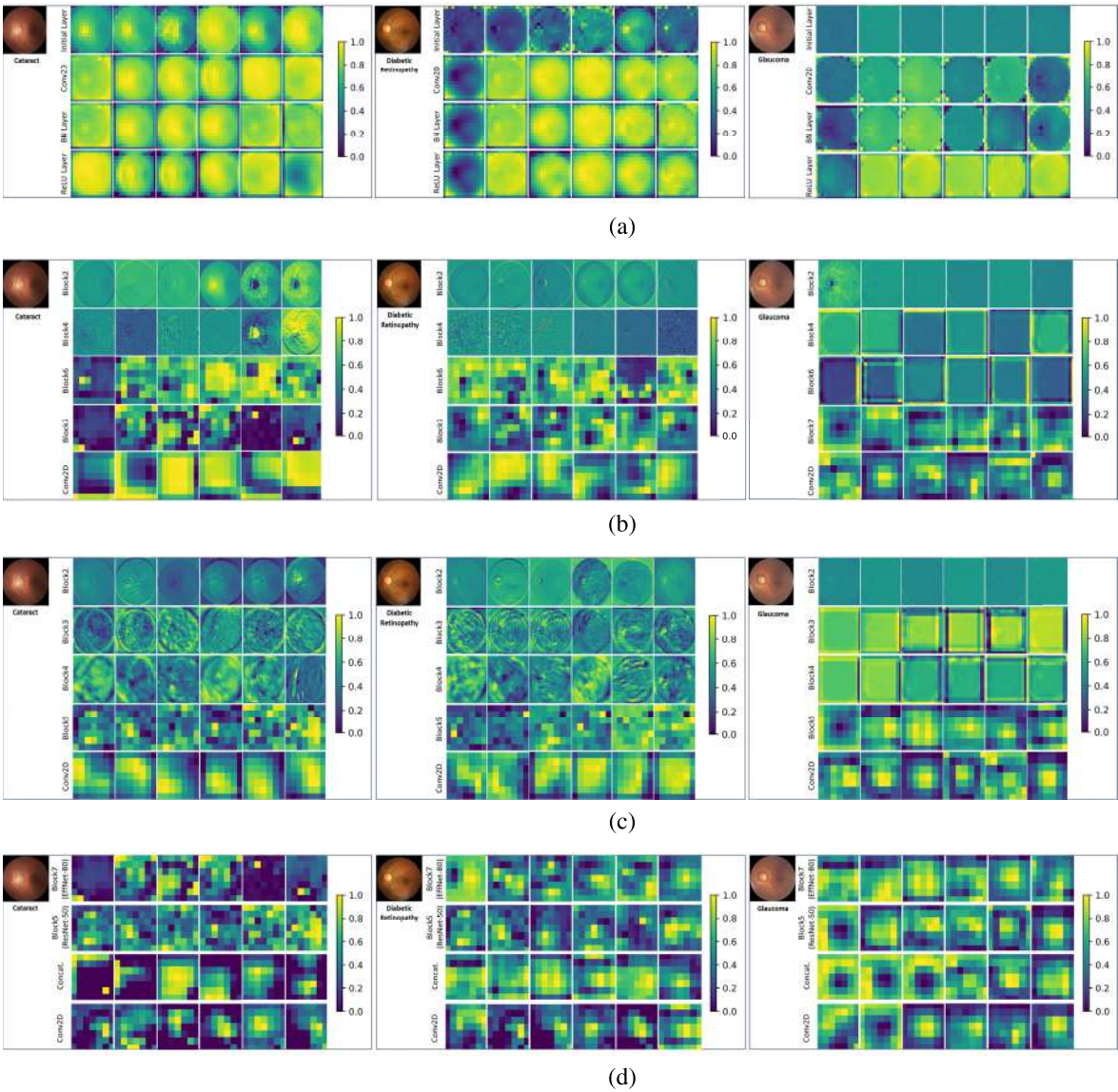


FIGURE 20. Features of the proposed models on the retinal dataset: (a) ViT, (b) EfficientNetB0-ViT, (c) ResNet50-ViT, and (d) EffResNet-ViT.

TABLE 7. Ablation study performance comparison of proposed models on the retinal dataset. The best performance is highlighted in bold.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Spec (%)	MCC (%)	AUC (%)
ViT w/o efB0 & R50	80.50	80.60	80.60	80.60	95.20	73.20	95.20
efB0 + ViT (no R50)	92.54	92.60	92.50	92.60	98.82	97.60	98.80
R50 + ViT (no efB0)	91.85	91.95	91.90	91.95	98.60	94.50	98.65
Proposed EffResNet-ViT (efB0 + R50 + ViT)	92.54	92.58	92.50	92.55	99.91	99.20	99.91

G. CLINICAL IMPACT

The EffResNet-ViT model represents a transformative advancement in automated diagnostic support for clinical settings, driven by its exceptional accuracy and refined feature extraction capabilities. Its integration allows clinicians to perform more precise and efficient analyses of medical images, specifically for BT and retinal disease diagnoses.

By providing highly reliable and accurate results, EffResNet-ViT contributes to reducing diagnostic errors, optimizing clinical workflows, and ultimately enhancing the quality of patient care.

Furthermore, EffResNet-ViT architecture promotes interpretability through visualizations such as t-SNE plots, providing clinicians with valuable insights into the image

regions that influence the model's classification decisions. This transparency fosters trust in AI-driven diagnoses by allowing clinicians to understand the underlying reasoning behind the model's predictions. The model's ability to generalize across a broad spectrum of tumor types and retinal conditions broadens its applicability in diverse healthcare environments, facilitating early and accurate diagnoses and supporting optimized treatment planning for a diverse range of patients.

VI. COMPARISON WITH SOTA METHODS

A. COMPARISON WITH THE BT CE-MRI DATASET

We conducted a detailed comparative analysis of the proposed EffResNet-ViT* model against several SOTA methods for BT CE-MRI dataset classification. The results are summarized in Table 8, revealing that the proposed model consistently outperformed the existing approaches in all key performance metrics.

Among SOTA methods, RanMerFormer [49] achieved a high accuracy of 98.86%, a precision of 98.87%, a sensitivity of 98.46% and a specificity of 99.39%. However, our proposed EffResNet-ViT surpasses RanMerFormer in all metrics, achieving an exceptional accuracy of 99.31%, precision of 99.26%, recall of 99.25%, F1-score of 99.26%, sensitivity of 99.25%, and specificity of 99.78%. Moreover, our model achieved a higher MCC of 99.08% and AUC of 99.91%, demonstrating its robustness in handling complex BT classification tasks compared with SOTA methods.

Compared with other SOTA methods, the superiority of EffResNet-ViT* becomes more evident. ResViT [50] achieved a high accuracy of 98.47%, precision 98.45%, recall 98.61%, and F1 score 98.53%; however, it falls short in specificity, MCC, and AUC compared with our proposed model. Similarly, the ensemble of ViT [51] achieved an accuracy of 98.70%, sensitivity of 97.78%, and specificity of 99.42%; however, it lacked comprehensive reporting of metrics such as precision, Recall, F1-score, MCC, and AUC, limiting its holistic evaluation. The hybrid DenseNet-121+ResNet-50+MLP [52] achieved an AUC of 99.87%, which was close to the 99.91% AUC of EffResNet-ViT*. However, it outperformed all other metrics, including the accuracy, precision, recall, and F1-score. ARiViT [53] also performed well, with an accuracy of 98.30% and precision of 98.31%; however, its lack of specificity, MCC, and AUC metrics further underscores the advantage of our model's comprehensive evaluation. Additionally, the EDN-SVM [54] model achieved an accuracy of 97.93%, sensitivity of 92.00%, and specificity of 98.00%, which are significantly lower than those achieved by EffResNet-ViT*.

Our analysis also highlights the limitations of the baseline ViT model, which achieved only 80.09% accuracy, demonstrating the inadequacy of pure transformer architectures in capturing both local and global features in complex datasets. While the EfficientNetB0-ViT model improved over the

baseline with an accuracy of 98.32%, the ResNet50-ViT model also demonstrated a strong performance with an accuracy, precision, recall, F1-score, sensitivity, specificity, MCC, and AUC of 95.58%, 95.65%, 95.58%, 95.55%, 95.58%, 98.53%, 94.10%, and 99.73%, respectively. However, the EffResNet-ViT* model achieved a significant improvement with 99.31% accuracy, surpassing all other models in terms of precision, recall, F1-score, sensitivity, specificity, MCC, and AUC. These results highlight the effectiveness of combining EfficientNet-B0 and ResNet-50 backbones with ViT to enhance feature extraction and classification performance.

The proposed EffResNet-ViT* model not only achieves the highest scores across all reported metrics but also demonstrates superior robustness and generalization capability, surpassing all previous SOTA methods. This exceptional performance of EffResNet-ViT* makes it a reliable and advanced solution for BT classification, addressing critical challenges in medical image analysis.

B. COMPARISON WITH THE RETINAL DATASET

Here, we comprehensively compare the proposed EffResNet-ViT* model with SOTA models on the retinal dataset. The results are summarized in Table 9. The proposed EffResNet-ViT* model demonstrates significant improvements across all key performance metrics compared with existing approaches.

A comparative analysis of the proposed EffResNet-ViT model against SOTA methods on the retinal dataset reveals significant improvements in performance across multiple metrics. The EffResNet-ViT model achieves the highest accuracy, precision, recall, F1-score, sensitivity, specificity, MCC, and AUC of 92.54%, 92.60%, 92.54%, 92.55%, 92.54%, 97.52%, 90.05%, and 98.82%, respectively, outperforming all the other SOTA methods and baseline models. The results are summarized in Table 9.

Compared with traditional CNN architectures, EffResNet-ViT shows substantial improvements. It surpasses the performance of VGG-19 [57], which achieves 23.33% accuracy; ResNet-50 [57], which achieves 86.19% accuracy; and MobileNet-V2 [57], which achieves 59.04% accuracy with wide margins. Compared to more advanced architectures, such as Inception-V3 [57] and DenseNet-121 [57], both achieving 90.23% accuracy, the proposed model demonstrates superior performance.

EffResNet-ViT also outperforms hybrid approaches such as GM-Net+ResNet-50 [55], which achieves 72.00% accuracy, and GM-Net+VGG-16 [55], which achieves 70.00% accuracy. It even surpasses the fusion of GM-Net+PMNet [55], which achieved 90.00% accuracy, 93.00% sensitivity, 97.00% specificity, and 83.00% MCC.

The proposed model also shows superior performance compared to the semi-supervised GAN [58], which achieved 87.00% accuracy, 85.00% F1-score, 83.33% AUC, and ensemble-CNN [59], which achieved 89.00% accuracy, 63.00% precision, 58.00% recall, 89.00%, F1-score, and

TABLE 8. Comparison results of the proposed and SOTA models on the BT CE-MRI datasets. The bold values indicate high scores for the respective metrics.

Method	Metrics							
	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Sensitivity (%)	Specificity (%)	MCC (%)	AUC (%)
ResViT [50]	98.47	98.45	98.61	98.53	-	-	-	-
Ensemble of ViT [51]	98.70	-	-	-	97.78	99.42	-	-
DenseNet-121+ResNet-50+MLP [52]	97.71	97.71	97.71	97.70	-	-	-	99.87
EDN-SVM [54]	97.93	-	-	-	92.00	98.00	-	-
ARiViT [53]	98.30	98.31	98.30	98.28	-	-	94.93	-
RanMerFormer [49]	98.86	98.87	-	-	98.46	99.39	-	-
ViT	80.09	80.31	80.09	79.49	80.09	93.26	73.53	95.49
EfficientNetB0-ViT	98.32	98.32	98.32	98.30	98.32	99.45	97.75	99.95
ResNet50-ViT	95.58	95.65	95.58	95.55	95.58	98.53	94.10	99.73
EffResNet-ViT*	99.31	99.26	99.25	99.26	99.25	99.78	99.08	99.91

TABLE 9. Comparison results of the proposed and SOTA models on the retinal datasets. The bold values indicate high scores for the respective metrics.

Method	Metrics							
	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Sensitivity (%)	Specificity (%)	MCC (%)	AUC (%)
GM-Net+ResNet-50 [55]	72.00	-	-	-	87.00	55.00	58.00	-
GM-Net+VGG-16 [55]	70.00	-	-	-	63.00	41.00	59.00	-
GM-Net+PMNet [55]	90.00	-	-	-	93.00	97.00	83.00	-
ResNet-50 [56]	-	-	-	92.00	-	-	-	-
VGG-19 [57]	23.33	-	-	-	-	-	-	-
ResNet-50 [57]	86.19	-	-	-	-	-	-	-
MobileNet-V2 [57]	59.04	-	-	-	-	-	-	-
Inception-V3 [57]	90.23	-	-	-	-	-	-	-
DenseNet-121 [57]	90.23	-	-	-	-	-	-	-
EfficientNet-B0 [57]	90.95	-	-	-	-	-	-	-
Semi-Supervised GAN [58]	87.00	-	-	85.00	-	-	-	83.33
Ensemble-CNN [60]	89.00	63.00	58.00	89.00	-	-	-	73.00
ViT	80.09	80.31	80.09	79.49	80.09	93.26	73.53	95.49
EfficientNetB0-ViT	98.32	98.32	98.32	98.30	98.32	99.45	97.75	99.95
ResNet50-ViT	95.58	95.65	95.58	95.55	95.58	98.53	94.10	99.73
EffResNet-ViT*	99.31	99.26	99.25	99.26	99.25	99.78	99.08	99.91

73.00% AUC. This demonstrates the effectiveness of the hybrid CNN-transformer architecture in capturing both local and global features in retinal images.

Among the proposed variants, EffResNet-ViT outperforms ViT, EfficientNetB0-ViT, and ResNet50-ViT across all metrics, indicating that combining EfficientNet-B0 and

ResNet-50 architectures with ViT yields the most effective model for retinal image classification.

The high specificity of 97.52% and sensitivity of 92.54% of the EffResNet-ViT model suggest its robust performance in correctly identifying both positive and negative cases, which is crucial for medical image analysis. The high AUC

of 98.82% of the model confirms its excellent discriminative ability across different retinal conditions.

The proposed EffResNet-ViT model demonstrates SOTA performance on the retinal dataset, outperforming existing SOTA methods and showcasing the potential of hybrid CNN-transformer architectures in medical image analysis tasks.

VII. CONCLUSION

In this study, we propose EFFResNet-ViT, a hybrid DL model that combines the strengths of EfficientNet-B0 and ResNet-50 with the ViT model to enhance the performance and interpretability of medical image classification. The proposed model achieves SOTA results on the BT CE-MRI and retinal datasets with accuracy rates of 99.31% and 92.54%, respectively. The model provides interpretability through feature fusion and incorporation of Grad-CAM visualizations, shedding light on regions that contribute to classification decisions, which is an essential aspect for clinical adoption.

Additionally, t-SNE visualizations of the model's embeddings on both datasets demonstrate clear class separation, indicating the effectiveness of EFFResNet-ViT in distinguishing between various tumor types and retinal conditions. The high clarity observed in the t-SNE plots supports the capacity of the model to learn discriminative features and generalize well across different classes.

Although the proposed model shows superior performance, future work can explore the impact of further hyperparameter tuning, different network architectures, and the inclusion of additional data sources, such as multimodal medical imaging, to further improve the robustness and generalization. Nevertheless, the ability of EFFResNet-ViT to provide both high performance and explainability presents a promising approach for automated and interpretable medical image analysis.

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